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DRIVE UNIT AND ELECTRIC VEHICLE

Abstract

A drive unit used for an electric vehicle includes: a motor; a crankshaft; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit the power to the sprocket, wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and at least part of the gear is disposed inside the opening portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The entire disclosure of Japanese Patent Application No. 2024-035841 filed on Mar. 8, 2024 including the specification, claims, drawings, and abstract is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a drive unit, and an electric vehicle that includes the drive unit.

BACKGROUND

[0003] Conventionally, as a drive unit for use in an electric vehicle, there has been a known drive unit that includes a motor, and a power transmission mechanism for transmitting a rotational power of the motor to a sprocket (for example, see JP 7246001 B).

SUMMARY

Technical Problem

[0004] In recent years, there has been demand for reduction in the size of drive units. In JP 7246001 B, a motor included in the drive unit and a board connected to the motor are placed close together to reduce the size of a drive unit. Meanwhile, the prior arts, including JP 7246001 B, leave room for improvement in reducing the size of the drive unit.

Solution to Problem

[0005] A drive unit of an aspect of the present disclosure is a drive unit for use in an electric vehicle, including: a motor that includes a rotor having an opening portion at a rotation center; a housing that houses the motor; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit to the sprocket, wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and at least part of the gear is disposed inside the opening portion.

[0006] A drive unit of an aspect of the present disclosure is a drive unit for use in an electric vehicle, including: a motor that includes a rotor having an opening portion at a rotation center; a housing that houses the motor; a crankshaft that is rotatably attached to the housing, and is rotatable by a human-power driving force; a first sprocket that outputs a rotational power of the motor; a second sprocket that outputs a rotational power of the human-power driving force; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit to the first sprocket, wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and at least part of the gear is disposed inside the opening portion.

[0007] A drive unit of an aspect of the present disclosure is a drive unit for use in an electric vehicle, including: a motor that includes a rotor having an opening portion at a rotation center; a housing; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit to the sprocket, wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and a rotation axis of the rotor and a rotation axis of the crankshaft are concentrically disposed.

[0008] A drive unit of an aspect of the present disclosure is a drive unit for use in an electric vehicle, including: a motor; a housing that houses the motor; a crankshaft rotatably attached to the

housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit to the sprocket; and a rotation detecting portion that detects rotation of the crankshaft or of a member rotating integrally with the crankshaft, wherein the rotation detecting portion rotates integrally with the crankshaft.

[0009] A drive unit of an aspect of the present disclosure is a drive unit for use in an electric vehicle, including: a motor; a housing that houses the motor; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit to the sprocket, wherein the motor includes: a motor shaft that is an output shaft; a stator fixed to the housing; and a rotor that is fixed to the motor shaft and includes a magnet, the rotor is disposed so as to face the stator in a direction along a rotation axis of the motor shaft, the power transmission mechanism includes at least one gear provided with tooth portions on a surface, and the stator and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft.

Advantageous Effects of Invention

[0010] The drive unit according to the present disclosure can be of reduced size. Despite its small size, the drive unit according to the present disclosure can produce sufficient output characteristics for a drive source of an electric vehicle.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] Embodiments of the present disclosure will be described based on the following figures, wherein:

[0012] FIG. 1 shows an appearance of a power-assisted bicycle according to a first embodiment;

[0013] FIG. 2 is a sectional view of a drive unit according to the first embodiment;

[0014] FIG. 3 is a sectional view of the drive unit according to the first embodiment, and shows an enlarged view of the vicinity of a motor;

[0015] FIG. 4 is a sectional view of a one-way clutch included in the drive unit according to the first embodiment;

[0016] FIG. 5 is a sectional view of a drive unit according to a second embodiment;

[0017] FIG. 6 is a sectional view of the drive unit according to the second embodiment, and shows an enlarged view of the vicinity of a motor;

[0018] FIG. 7 is a sectional view of a drive unit according to a third embodiment;

[0019] FIG. 8 is a sectional view of the drive unit according to the third embodiment, and shows an enlarged view of the vicinity of a motor;

[0020] FIG. 9 is a sectional view of a drive unit according to a fourth embodiment;

[0021] FIG. 10 is a sectional view of the drive unit according to the fourth embodiment, and shows an enlarged view of the vicinity of a motor;

[0022] FIG. 11 is a sectional view of a drive unit according to a fifth embodiment;

[0023] FIG. 12 is a sectional view of the drive unit according to the fifth embodiment, and shows an enlarged view of the vicinity of a motor;

[0024] FIG. 13 is a sectional view of a drive unit according to a sixth embodiment;

[0025] FIG. 14 is a sectional view of a drive unit according to a seventh embodiment;

[0026] FIG. 15 is a sectional view of the drive unit according to the seventh embodiment, and shows an enlarged view of the vicinity of a motor;

[0027] FIG. 16 is a sectional view of a drive unit according to an eighth embodiment;

[0028] FIG. 17 is a sectional view of the drive unit according to the eighth embodiment, and shows an enlarged view of the vicinity of a motor;

[0029] FIG. **18** is a sectional view of a drive unit according to a ninth embodiment;
[0030] FIG. **19** is a sectional view of a drive unit according to a tenth embodiment;
[0031] FIG. **20** is a sectional view of a drive unit according to an eleventh embodiment;
[0032] FIG. **21** is a sectional view of the drive unit according to the eleventh embodiment, and shows an enlarged view of the vicinity of a motor;
[0033] FIG. **22** is a sectional view of a drive unit according to a twelfth embodiment; and
[0034] FIG. **23** is a sectional view of the drive unit according to the twelfth embodiment, and shows an enlarged view of the vicinity of a motor.

DESCRIPTION OF EMBODIMENTS

[0035] Hereinafter, a drive unit, and an electric vehicle that includes the drive unit according to the present disclosure are described in detail. Each of embodiments described below is only an example, and the present disclosure is not limited to the following embodiments. Any forms made up by selectively combining the embodiments and modification examples described below are encompassed in the present disclosure.

First Embodiment

[0036] Hereinafter, by reference to FIGS. **1** to **4**, a first embodiment of a drive unit and an electric vehicle in the present disclosure is described.

[0037] FIG. **1** is a right side view showing an appearance of a power-assisted bicycle **1** that is an electric vehicle according to the first embodiment. Note that the electric vehicle in the present disclosure is not limited to the power-assisted bicycle **1**, and may be an electric vehicle that travels by the power of a motor. The power-assisted bicycle **1** in the present disclosure is not limited to a sport bicycle as shown in FIG. **1**, and may be, for example, a city bike, a folding bicycle or the like. Hereinafter, the front-rear, left-right, and up-down directions in the embodiments in the present disclosure mean the front-rear, left-right, and up-down directions with reference to a state in which a rider is seated on a saddle **5** of the power-assisted bicycle **1** facing handlebars **4**.

[0038] As shown in FIG. **1**, similar to a typical bicycle, the power-assisted bicycle **1** includes a frame **2**, a front wheel **3A**, a rear wheel **3B**, the handlebars **4**, and the saddle **5**.

[0039] The frame **2** is a framework that connects the front wheel **3A**, the rear wheel **3B**, the handlebars **4**, the saddle **5**, and the like. The frame **2** includes a plurality of tubes. A head tube **2A**, front forks **2B**, a top tube **2C**, a bottom tube **2D**, a seat tube **2E**, seat stays **2F**, and chain stays **2G** are provided as the plurality of tubes.

[0040] The power-assisted bicycle **1** further includes a drive unit **20**, and a power storage device **10** that supplies the drive unit **20** with electric power. The power-assisted bicycle **1** has a function of assisting the driving force resulting from the pedaling force of a rider (human-power driving force) with the power of a motor **40** (see FIG. **2** described later) of the drive unit **20**.

[0041] As described in detail later, part of the drive unit **20** is housed in the bottom tube **2D**. Note that the drive unit **20** may be attached to the outside of the frame **2**, instead of being housed in the bottom tube **2D**.

[0042] The drive unit **20** includes a pair of crank arms **6** to which pedals **7** are respectively attached. The crank arms **6** are provided to the left and right of the power-assisted bicycle **1**, one for each side. The other end portions of the pair of crank arms **6** are both coupled to a crankshaft **30** (see FIG. **2**).

[0043] The drive unit **20** includes a sprocket **70**. The power-assisted bicycle **1** includes a chain **8** as a mechanism that transmits the rotational power of the sprocket **70** to the rear wheel **3B**. The chain **8** couples the sprocket **70** to a rear wheel sprocket **11** provided to the rear wheel **3B**. Note that the sprocket **70** and the rear wheel sprocket **11** may be coupled by a belt or shaft drive.

[0044] The power storage device **10** is a power source device that supplies the drive unit **20** with electric power. In an example shown in FIG. **1**, the power storage device **10** is attached outside the bottom tube **2D**. Note that the position of the power storage device **10** is not limited to the bottom tube **2D**, and this device may be attached to the top tube **2C** or the seat tube **2E**. Alternatively, the

power storage device **10** may be housed in the frame **2**.

[0045] Preferably, a secondary battery is used as a battery housed in the power storage device **10**. Alternatively, a primary battery may be used. The voltage of the power storage device **10** may range, for example, from 12 V to 48 V, inclusive. The power storage device **10** may be provided with a voltage conversion circuit. The voltage conversion circuit may be provided in the power storage device **10**, or provided outside the power storage device **10**.

[0046] Next, by reference to FIG. **2**, the drive unit **20** according to the first embodiment is described. FIG. **2** is a sectional view of the drive unit **20**.

[0047] As shown in FIG. **2**, the drive unit **20** includes: the motor **40** for applying rotational power for assisting the pedaling force on the pedals **7** (see FIG. **1**); a housing **31** that houses the motor **40**; and the crankshaft **30** rotatably attached to the housing **31**. The drive unit **20** is a so-called center unit type drive unit.

[0048] The drive unit **20** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40**; the sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50** that decelerates the rotational power of the motor **40**, and transmits it to the sprocket **70** via the output body **62**.

[0049] As described in detail later, the output body **62** functions as a power combination body at which the rotational power of the human-power driving force and the rotational power of the motor **40** are combined. In the present embodiment, the power transmission mechanism **50** is made up of two stages of speed reduction mechanisms, and includes: a planetary gear mechanism (first power transmission mechanism) **51** that is a first-stage speed reduction mechanism; and a second speed reduction mechanism (second power transmission mechanism) **52** that is a second-stage speed reduction mechanism.

[0050] As shown in FIG. **2**, the motor **40** includes a stator **41**, a rotor **42**, and a motor shaft **43** that rotates integrally with the rotor **42**. In the present embodiment, the motor **40** is an inner-rotor type motor. As described above, the motor **40** is housed in the housing **31**.

[0051] The stator **41** is fixed to the inside of the housing **31** by press-fitting or with adhesive or a pin. The rotor **42** has an annular shape that has a circular opening portion **44** (see FIG. **3**) at the rotation center, and is disposed inside the stator **41**. As described in detail later, a sun gear **511**, an internal gear **512**, planetary gears **513**, and a planetary gear shaft **514** that are gears constituting the planetary gear mechanism **51** are disposed inside the opening portion **44** in the radial direction.

[0052] The motor shaft **43** is the output shaft of the motor **40**, and is disposed such that the rotation axis is along the axial direction of the crankshaft **30**. The motor shaft **43** is rotatably supported by bearings **320**, **321**, and **322** provided in the housing **31**, so as to be rotatable with respect to the housing **31**. The rotor **42** and the motor shaft **43** are fixed to each other by serrations, for example.

[0053] The housing **31** is a member that constitutes an outer shell of the drive unit **20**. The housing **31** is formed mainly of metal, such as aluminum, stainless steel, or magnesium. Alternatively, a non-metal material may be used. No specific limitation is imposed on the material of the housing **31**.

[0054] The housing **31** includes: a first divided body **311** that constitutes the right half of the housing **31**; and a second divided body **312** that constitutes the left half of the housing **31**. The first divided body **311** and the second divided body **312** are joined together with a fastener member **34**, such as a bolt. By joining the first divided body **311** and the second divided body **312** to each other, the hollow housing **31** is formed. Note that no specific limitations are imposed on the size, shape, thickness, and the like of the housing **31**. The space formed in the housing **31** may be closed, or not closed.

[0055] The first divided body **311** includes a protruding portion **3111** that protrudes to the left. The internal gear **512** included in the planetary gear mechanism **51** is fitted to the protruding portion **3111**.

[0056] The human power transmission body **61**, part of the output body **62**, the planetary gear mechanism **51**, and the second speed reduction mechanism **52**, in addition to the motor **40**, are housed in the housing **31**. The housing **31** is provided with through-holes **314** and **315** through which the crankshaft **30** penetrates, at its opposite ends.

[0057] In the present embodiment, part of the housing **31** is housed in the bottom tube **2D**. Between the housing **31** and the bottom tube **2D** (frame **2**) there is disposed a heat dissipation member **33** that transfers the heat of the housing **31** to the bottom tube **2D**. That is, the heat of the housing **31** is transferred to the bottom tube **2D** via the heat dissipation member **33**. In general, the temperature of the housing **31** tends to be increased by the heat of the motor **40** and the like housed in the housing **31**. As in the present embodiment, by disposing the heat dissipation member **33** between the housing **31** and the bottom tube **2D**, the temperature of the housing **31** can be reduced, and the reliability of the drive unit **20** can be improved. Note that it may be configured so as to allow the heat of the housing **31** to be transferred to members other than the bottom tube **2D** via the heat dissipation member **33**.

[0058] For example, an elastic body that has a higher thermal conductivity than typical resin and is compressible can be used as the heat dissipation member **33**. An example of a preferable heat dissipation member **33** is a heat release sheet that includes flexible resin, and thermal conductive filler dispersed in the resin. The thickness of the heat dissipation member **33** ranges, for example, from 0.3 mm to 2 mm, inclusive. Note that a conventionally publicly known member may be used as the heat dissipation member **33**.

[0059] The heat dissipation member **33** may have adherence, and be adhesive to the surface or the like of the housing **31** or the bottom tube **2D**. In the case where the heat dissipation member **33** is adhesive to the surface or the like of the housing **31** or the bottom tube **2D**, the heat dissipation member **33** can be easily installed. Note that the heat dissipation member **33** may be pasted to the surface or the like of the housing **31** or the bottom tube **2D** using adhesive, adhesive tape, or the like.

[0060] The crankshaft **30** is a cylindrical member rotationally driven by a human-power driving force. The crankshaft **30** rotates with respect to the drive unit **20**. In the present embodiment, the crankshaft **30** is made up of a hollow member, but may be made up of a solid member instead.

[0061] The opposite ends of the crankshaft **30** protrude outward from the housing **31**. A pair of crank arms **6** (see FIG. **1**) are respectively provided at the opposite ends of the crankshaft **30**. The crankshaft **30** is rotatably supported by bearings **324** and **325** at the housing **31** (frame **2**). The bearings **324** and **325** are made up of, for example, ball bearings.

[0062] The power transmission mechanism **50** is a mechanism that decelerates the rotational power of the motor **40**, and transmits the rotational power of the motor **40** to the sprocket **70** via the output body **62**. In the present embodiment, the power transmission mechanism **50** is made up of the two stages of the speed reduction mechanisms, and includes: the planetary gear mechanism **51** that is the first-stage speed reduction mechanism; and the second speed reduction mechanism **52** that is the second-stage speed reduction mechanism. Note that the power transmission mechanism **50** may include an acceleration mechanism.

[0063] As described in detail later, the planetary gear mechanism **51** includes: the motor shaft **43** as the sun gear **511**; the internal gear **512** disposed on a concentric circle of the sun gear **511**; the planetary gears **513** that mesh with the sun gear **511** and the internal gear **512**; the planetary gear shafts **514**; and a planetary carrier **515** as an output shaft. The internal gear **512** is non-rotatably fixed to the housing **31** by being fitted to the protruding portion **3111** of the housing **31**. The rotational power of the motor **40** output from the planetary carrier **515** is transmitted to the second speed reduction mechanism **52** via a one-way clutch **53**.

[0064] When the rotational power in a direction (hereinafter, called “forward direction”) of accelerating the power-assisted bicycle **1** in the traveling direction is applied to the planetary carrier **515**, the one-way clutch **53** transmits the rotational power to a first transmission gear **521** included

in the second speed reduction mechanism **52**. When the rotational power in the direction opposite to the forward direction is applied to the planetary carrier **515**, the one-way clutch **53** does not transmit the rotational power to the first transmission gear **521**. When the rotational power in the forward direction is applied to the first transmission gear **521**, the one-way clutch **53** does not transmit the rotational power to the planetary carrier **515**.

[0065] The second speed reduction mechanism **52** includes the first transmission gear **521**, and a second transmission gear **522**. The rotation axis of the first transmission gear **521** is positioned concentrically on the rotation axis of the sun gear **511**. The rotation axis of the second transmission gear **522** is positioned concentrically on the rotation axis of the crankshaft **30**. That is, the outer diameter of the second transmission gear **522** is larger than the outer diameter of the first transmission gear **521**. The number of teeth of the second transmission gear **522** is greater than the number of teeth of the first transmission gear **521**.

[0066] The first transmission gear **521** is disposed outside the planetary carrier **515** in the radial direction, and the rotational power of the motor **40** is transferred to this gear via the one-way clutch **53**. The first transmission gear **521** is rotatably supported by a bearing **323** (first bearing) at the housing **31**.

[0067] The second transmission gear **522** meshes with tooth portions provided on the outer peripheral surface of the output body **62**. Accordingly, the rotational power of the motor **40** transmitted to the second transmission gear **522** is transmitted to the output body **62**.

[0068] Here, the bearing **323** (first bearing) that rotatably supports the first transmission gear **521**, and the one-way clutch **53** may be disposed at positions overlapping in the axial direction of the crankshaft **30**. Accordingly, reduction in size of the drive unit **20** can be facilitated.

[0069] The second speed reduction mechanism **52**, and a bearing **320** (second bearing) that is provided at the distal end (right end) of the motor shaft **43** as the sun gear **511** and rotatably supports the motor shaft **43** are disposed at positions overlapping in the axial direction of the crankshaft **30**. Accordingly, reduction in size of the drive unit **20** can be facilitated.

[0070] Note that in the present embodiment, the second speed reduction mechanism **52** includes the two gears made up of the first transmission gear **521** and the second transmission gear **522**.

However, the number of gears may be one, or three or more. The power transmission mechanism **50** may be a single-stage speed reduction mechanism, or a speed reduction mechanism having three or more stages.

[0071] The human power transmission body **61** is a cylindrical member that extends along the axial direction of the crankshaft **30**, and is disposed on the outer peripheral portion of the crankshaft **30** in the housing **31**. The human power transmission body **61** rotates integrally with the crankshaft **30**. In the present embodiment, the human power transmission body **61** is divided into a first human power transmission body **611** and a second human power transmission body **612**. Note that the human power transmission body **61** may be made up of a single member.

[0072] The first human power transmission body **611** is coupled to the crankshaft **30**. Splines or serrations to be fitted to the crankshaft **30** are formed in the inner peripheral surface of the first human power transmission body **611**. Note that a tooth missing part may be provided at a site where the first human power transmission body **611** and the crankshaft **30** fit each other. This allows positioning in the assembling direction, and facilitates assembly. The first human power transmission body **611** and the crankshaft **30** may be fitted together by press-fitting or with a screw.

[0073] The second human power transmission body **612** is disposed on the right side of the first human power transmission body **611** in the axial direction of the crankshaft **30**. The second human power transmission body **612** is coupled to the first human power transmission body **611**, and transmits the rotational power to the output body **62**.

[0074] In the present embodiment, splines or serrations to be fitted to the inner peripheral surface of the left end of the second human power transmission body **612** are formed in the outer peripheral surface of the right end of the first human power transmission body **611**. Accordingly, the first

human power transmission body **611** and the second human power transmission body **612** are coupled to each other.

[0075] No specific limitations are imposed on the lengths of the first human power transmission body **611** and the second human power transmission body **612** in the axial direction of the crankshaft **30**, so long as the rotational power by the human-power driving force can be transmitted to the output body **62**. For example, each of the lengths ranges from 5% to 40%, inclusive, of the length of the crankshaft **30** in the axial direction. In the present embodiment, the length of the first human power transmission body **611** in the axial direction of the crankshaft **30** is greater than the length of the second human power transmission body **612**. Note that the length of the first human power transmission body **611** in the axial direction of the crankshaft **30** may be shorter than the length of the second human power transmission body **612**.

[0076] The output body **62** is a cylindrical member that extends along the axial direction of the crankshaft **30**, and is disposed on the outer peripheral portion of the crankshaft **30**. The rotation axis of the output body **62** is concentrically positioned on the rotation axis of the crankshaft **30**.

[0077] The length of the output body **62** in the axial direction of the crankshaft **30** is shorter than the length of the crankshaft **30**, and ranges, for example, from 10% to 50%, inclusive, of the length of the crankshaft **30** in the axial direction. The right end of the output body **62** protrudes to the outside of the housing **31** through the through-hole **314** provided in the housing **31**. The output body **62** is rotatably supported by the bearing **324** at the housing **31**.

[0078] Tooth portions to be meshed with the tooth portions of the second transmission gear **522** are provided on the outer peripheral surface of the output body **62**. Accordingly, the rotational power of the motor **40** is transmitted to the output body **62** via the second transmission gear **522**.

[0079] In the present embodiment, a one-way clutch **63** is provided between the second human power transmission body **612** and the output body **62**. When the rotational power in the forward direction is applied to the second human power transmission body **612**, the one-way clutch **63** transmits the rotational power to the output body **62**. When the rotational power in the direction opposite to the forward direction is applied to the second human power transmission body **612**, the one-way clutch **63** does not transmit the rotational power to the output body **62**. When the rotational power in the forward direction is applied to the output body **62** via the power transmission mechanism **50**, the one-way clutch **63** does not transmit the rotational power to the second human power transmission body **612**. The one-way clutch **63** also functions as a bearing, and rotatably supports the output body **62**. Note that the configuration of the one-way clutch **63** is described later.

[0080] Splines or serrations to be fitted to the sprocket **70** are formed in a portion of the output body **62** that protrudes to the outside of the housing **31**. Accordingly, the sprocket **70** rotates integrally with the output body **62**. Note that an elastic member for regulating the movement of the sprocket **70** may be disposed between the output body **62** and the sprocket **70**.

[0081] Note that the human power transmission body **61** and the output body **62** are provided separately in the present embodiment, but may be integrally configured instead. For example, the human power transmission body **61** is not necessarily provided, and it may be the case that only the output body **62** is provided. In this case, by transmitting the rotational power of the crankshaft **30** due to the human-power driving force to the output body **62**, the power-assisted bicycle **1** is allowed to travel forward.

[0082] A one-way clutch may be disposed between the output body **62** and the sprocket **70**. When the rotational power in the forward direction is applied to the output body **62**, the one-way clutch transmits the rotational power to the sprocket **70**. When the rotational power in the direction opposite to the forward direction is applied to the output body **62**, the one-way clutch does not transmit the rotational power to the sprocket **70**.

[0083] Here, the power transmission path in the drive unit **20** is described. First, the power transmission path of the human-power driving force applied to the pedals **7** is described. The rider

pushes down the pedals **7** of the power-assisted bicycle **1**, which rotates the crankshaft **30** in the forward direction. When the crankshaft **30** rotates in the forward direction, the human power transmission body **61** fitted to the outer peripheral portion of the crankshaft **30** rotates in the forward direction integrally with the crankshaft **30**. The rotational power of the human power transmission body **61** rotating in the forward direction is transmitted to the output body **62** for outputting the rotational power of the motor **40**. The output body **62**, and the sprocket **70** fixed to the output body **62** integrally rotate in the forward direction. When the sprocket **70** rotates in the forward direction, the rotational power in the forward direction is applied to the rear wheel sprocket **11** via the chain **8**, and the rear wheel sprocket **11** and the rear wheel **3B** rotate in the forward direction. Accordingly, the power-assisted bicycle **1** travels forward.

[0084] Next, the power transmission path of the rotational power of the motor **40** is described. When the motor shaft **43** of the motor **40** rotates in the forward direction, the rotational power is transmitted to the output body **62** via the power transmission mechanism **50**, which includes the planetary gear mechanism **51** and the second speed reduction mechanism **52**. That is, the output body **62** functions as a power combination body at which there are combined the rotational power from the human power transmission body **61** by the human-power driving force described above, and the rotational power of the motor **40**. The rotational power of the motor **40** transmitted to the output body **62** is transmitted to the rear wheel **3B** on the power transmission path similar to that for the human-power driving force described above, and the power-assisted bicycle **1** travels forward.

[0085] The drive unit **20** includes torque detecting portions **80** that detect the torque of the human-power driving force; and a control board **81** on which there is disposed a control device (not shown) for controlling the output of the motor **40** in response to the detected torque and achieving an appropriate power assisting function. The drive unit **20** further includes a rotation detecting portion **82** for detecting the rotation of the crankshaft **30**.

[0086] The torque detecting portions **80** are provided on the outer peripheral portion of the first human power transmission body **611**, and detect the strain applied to the first human power transmission body **611**. For example, a strain sensor that detects a physical strain, or a strain sensor that detects a magnetic strain may be used as each torque detecting portion **80**. For example, a scheme may be used in which a paint that emits light in response to a mechanical stimulus is applied to the outer peripheral portion of the human power transmission body **61** and the light is detected by an optical sensor.

[0087] In the present embodiment, the drive unit **20** includes total two torque detecting portions **80**, one at each of the opposite positions in the circumferential direction on the outer peripheral portion of the first human power transmission body **611**. Note that the number of torque detecting portions **80** provided on the outer peripheral portion of the first human power transmission body **611** may be only one, or three or more. The installation positions of the torque detecting portions **80** are not limited to the outer peripheral portion of the first human power transmission body **611**, and may be on, for example, the outer peripheral portion of the second human power transmission body **612** or the crankshaft **30**.

[0088] The control device is disposed on the control board **81**, and controls the rotation of the motor **40** based on the torque detected by each torque detecting portion **80**. The control device controls the rotation of the motor **40** based on detection information on the rotation detecting portion **82**. A conventionally publicly known configuration may be adopted as the configuration of the control device.

[0089] The torque detecting portion **80** and the rotation detecting portion **82** may communicate with the control device by means of signal lines or wireless signals. In the case of communication by means of the wireless signals, cables that connect the torque detecting portion **80** and the rotation detecting portion **82** to the control device are not required, which facilitates reduction in size of the drive unit **20**.

[0090] In the present embodiment, the control board **81** is made up of a single board that has through-holes through which the crankshaft **30** and the motor **40** penetrate. The control board **81** is disposed such that the substantially normal direction of the surface (thickness direction) can be along the axial direction of the crankshaft **30**. Note that the control board **81** may be made up of multiple boards.

[0091] Part of the control board **81** is joined to the motor **40**. The part of the control board **81** is joined to the motor **40**, and the control board **81** and the motor **40** are disposed close to each other, which can shorten the cable that connects the control board **81** to the motor **40**, thus facilitating reduction in the size of the drive unit **20**.

[0092] Part of the control board **81** is joined to the second divided body **312**. In the present embodiment, the control board **81** and the second divided body **312** are joined to each other via a heat dissipation member **35**. A configuration similar to that of the heat dissipation member **33** intervening between the housing **31** and the down tube **2D** may be used for the heat dissipation member **35**.

[0093] The control board **81** may be provided with, for example, a rotor rotation detecting portion **85** for detecting the rotation of the rotor **42**. The rotor rotation detecting portion **85** includes, for example, a Hall element. The Hall element detects the variation in the magnetic field caused by the rotation of the rotor **42**, thus allowing measurement of the number of rotations of the rotor **42**. Note that the magnetic force detected by the Hall element may be the magnetic force of a member that rotates together with the rotor **42**.

[0094] A cable **83** extending from the outside of the drive unit **20** is connected via a connector **84** to the control board **81**. Note that the connector **84** is an element separated from the drive unit **20**. The cable **83** is connected to, for example, the power storage device **10** (see FIG. 1), and supplies electric power to electronic devices disposed on the control board **81**.

[0095] The connector **84** includes a first portion **841** that extends in a substantially normal direction of a surface of the control board **81**, and a second portion **842** that extends in a direction inclined from the substantially normal direction of the surface of the control board **81**. In the present embodiment, the second portion **842** is configured so as to be substantially perpendicular to the first portion **841**. That is, the connector **84** is bent substantially perpendicularly. Accordingly, the length of the connector **84** in the left-right direction can be reduced, which can reduce the size of the drive unit **20**.

[0096] The rotation detecting portion **82** detects the number of rotations of the crankshaft **30**. The rotation detecting portion **82** internally includes, for example, a Hall element, and is disposed at a position overlapping the magnet **86** provided on the outer peripheral portion of the first human power transmission body **611**. The Hall element detects the variation in the magnetic field of the magnet by the rotation of the crankshaft **30**, thus allowing measurement of the number of rotations of the crankshaft **30**. Note that the configuration of the rotation detecting portion **82** is not limited to this. The rotation detecting portion **82** may be, for example, an inertial sensor, such as an acceleration sensor or a gyroscope sensor.

[0097] In the present embodiment, the rotation detecting portion **82** is disposed on the control board **81**. By disposing the rotation detecting portion **82** on the control board **81**, the rotation detecting portion **82** can be disposed without increasing the number of boards. Accordingly, reduction in size of the drive unit **20** can be facilitated.

[0098] Next, by further reference to FIG. 3, the motor **40**, and the gears constituting the planetary gear mechanism **51** are described in detail. FIG. 3 is an enlarged view of the vicinity of the motor **40** in FIG. 2.

[0099] As shown in FIG. 3, the motor **40** includes the stator **41**, the rotor **42**, and the motor shaft **43** that rotates integrally with the rotor **42**.

[0100] The stator **41** is fixed to the inside of the housing **31** by press-fitting, with adhesive, by screwing, or with a pin. In this case, an elastic member, such as any of a rubber member and

urethane, may be provided between the stator **41** and the housing **31**. By providing the elastic member, vibrations of the stator **41** can be reduced. A heat release sheet or heat release paste (heat release grease, heat release gel, etc.) may be provided between the stator **41** and the housing **31**. Thus, the heat dissipation property of the motor **40** can be improved, and the motor **40** can be prevented from overheating and vibrating.

[0101] Preferably, the outer diameter of the stator **41** is equal to or less than 100 mm. More preferably, the outer diameter is equal to or less than 90 mm. By making the stator **41** have an outer diameter of 100 mm or less, reduction in the size of the drive unit **20** can be further facilitated. The lower limit of the outer diameter of the stator **41** is, for example, 60 mm.

[0102] The rotor **42** has an annular shape that has a circular opening portion **44** at the rotation center, and is disposed inside the stator **41**. Preferably, the inner diameter of the rotor **42** is equal to or larger than 30 mm. More preferably, the inner diameter is equal to or larger than 40 mm. By making the rotor **42** have an inner diameter of 30 mm or more, arrangement of the gears constituting the planetary gear mechanism **51** inside the opening portion **44** of the rotor **42** in the radial direction is facilitated. The upper limit of the inner diameter of the rotor **42** is, for example, 100 mm.

[0103] The motor shaft **43** is a cylindrical shaft-shaped member extending along the axial direction of the crankshaft **30** (left-right direction), and rotates integrally with the rotor **42**. The motor shaft **43** is disposed such that its rotation axis is along the axial direction of the crankshaft **30**. The motor shaft **43** is rotatably supported by the bearings **320**, **321**, and **322** provided in the housing **31**, so as to be rotatable with respect to the housing **31**.

[0104] The rotor **42** and the motor shaft **43** are fixed to each other by serrations. Note that the rotor **42** and the motor shaft **43** may be fixed by press-fitting, shrink-fitting, cool-fitting, or the like. When the rotor **42** and the motor shaft **43** are fixed, an elastic member, such as any of a rubber member and urethane, may be provided between the rotor **42** and the motor shaft **43**.

[0105] The motor shaft **43** is disposed such that the central axis of rotation passes through the center of the internal gear **512** included in the planetary gear mechanism **51**, described later, and penetrates through the internal gear **512**. Tooth portions **43A** to mesh with tooth portions **513A** of the planetary gears **513** included in the planetary gear mechanism **51** are provided on the surface of the motor shaft **43**. The motor shaft **43** constitutes the sun gear **511** of the planetary gear mechanism **51**. Note that the tooth portions in this specification indicate tooth shapes provided in the surfaces of the gears, and encompass areas that do not mesh with the tooth portions of other gears.

[0106] As shown in FIG. 3, the planetary gear mechanism **51** includes: the motor shaft **43** as the sun gear **511**; the internal gear **512** disposed on the concentric circle of the sun gear **511**; the planetary gears **513** that mesh with the sun gear **511** and the internal gear **512**; the planetary gear shafts **514**; and the planetary carrier **515** as the output shaft.

[0107] The internal gear **512** has an annular shape. Tooth portions **512A** are formed over the entire region of the inner peripheral surface. The internal gear **512** is non-rotatably fixed to the housing **31** by being fitted to the protruding portion **3111** of the housing **31**.

[0108] The tooth portions **513A** are formed over the entire periphery of each planetary gear **513**, which meshes with the sun gear **511** and the internal gear **512**. The planetary gear shafts **514** penetrate through the centers of the respective planetary gears **513**, and freely rotatably hold these planetary gears **513**. A bush **38** having L-shape in a sectional view taken in the axial direction is disposed outside of each planetary gear shaft **514** in the radial direction, and is configured such that the corresponding planetary gear shaft **514** can rotate. Note that the bearing **323** is not necessarily provided, and it may be the case that only the bush **38** is provided. The planetary carrier **515** supports the planetary gears **513** to allow them to orbit. The rotation axis of the planetary carrier **515** is positioned concentrically on the rotation axis of the sun gear **511**. The planetary carrier **515** is rotatably supported by the bearing **323** provided in the housing **31**. The bearing **323** is made up

of, for example, a ball bearing.

[0109] As shown in FIG. 3, the rotor **42**, and the tooth portions **43A** of the motor shaft **43** (sun gear **511**), the tooth portions **512A** of the internal gear **512**, and the tooth portions **513A** of the planetary gears **513** are disposed at overlapping positions in the axial direction of the crankshaft **30**.

Accordingly, in the axial direction of the crankshaft **30**, the size of the drive unit **20** can be reduced. Note that the tooth portions **43A** of the motor shaft **43** (sun gear **511**), the tooth portions **512A** of the internal gear **512**, and the tooth portions **513A** of the planetary gears **513** may be disposed at positions overlapping the stator **41**, instead of the rotor **42**, in the axial direction of the crankshaft **30**. Also in this case, the size of the drive unit **20** can be reduced.

[0110] At least parts of the tooth portions **43A** of the motor shaft **43**, the tooth portions **512A** of the internal gear **512**, and the tooth portions **513A** of the planetary gears **513** are disposed inside the opening portion **44** of the rotor **42** in the radial direction. Accordingly, the drive unit **20** can be further reduced in size.

[0111] Metal gears and resin gears may be used as the gears that constitute the planetary gear mechanism **51**. Use of resin gears can reduce the weight of the drive unit **20**, and furthermore, can alleviate noise during traveling. The resin material may be for example, polyacetal, nylon 46, nylon 66, PEEK, or thermosetting resin, or may be super engineering plastic.

[0112] The gears that constitute the planetary gear mechanism **51** may be spur gears or helical gears. Use of helical gears can alleviate noise during traveling. Preferably, the gears that constitute the planetary gear mechanism **51** are lubricated with grease. Note that the gears that constitute the planetary gear mechanism **51** may be lubricated with lubrication oil. Each gear may have a non-arc shape resembling a gear shape.

[0113] Next, by further reference to FIG. 4, the one-way clutch **63** is described. FIG. 4 is a sectional view of the one-way clutch **63**.

[0114] As shown in FIG. 2, the one-way clutch **63** is disposed between the second human power transmission body **612** and the output body **62**. The one-way clutch **63** is disposed closer to the sprocket **70** in the axial direction of the crankshaft **30** than the axial-direction center portion of the crankshaft **30**. As described above, when the rotational power in the forward direction is applied to the second human power transmission body **612**, the one-way clutch **63** transmits the rotational power to the output body **62**. When the rotational power in the direction opposite to the forward direction is applied to the second human power transmission body **612**, the one-way clutch **63** does not transmit the rotational power to the output body **62**.

[0115] As shown in FIG. 4, the one-way clutch **63** includes an inner wheel body **63A**, and an outer wheel body **63B** that covers the outer peripheral portion of the inner wheel body **63A**. The inner wheel body **63A** is provided on the outer peripheral portion of the second human power transmission body **612**. Note that the inner wheel body **63A** may be provided integrally with the second human power transmission body **612**. The outer wheel body **63B** is provided on the inner peripheral portion of the output body **62**. Note that the outer wheel body **63B** may be formed integrally with the output body **62**.

[0116] Sprags **63C** as engagement pieces are disposed between the inner wheel body **63A** and the outer wheel body **63B**. That is, the one-way clutch **63** is a so-called sprag type one-way clutch. The sprags **63C** are oriented in a stretching-out attitude between the inner wheel body **63A** and the outer wheel body **63B**, thus allowing transmission of the rotational power. On the other hand, the sprags **63C** slide between the inner wheel body **63A** and the outer wheel body **63B**, thus preventing transmission of the rotational power. The shape of each sprag **63C** is not limited to the form shown in FIG. 4, and may have, for example, a cylindrical shape.

[0117] Balls **63D** as rolling bodies are further disposed between the inner wheel body **63A** and the outer wheel body **63B**. Consequently, when the outer wheel body **63B** is rotatable with respect to the inner wheel body **63A** i.e., when the rotational power cannot be transmitted, the one-way clutch **63** receives the radial load, functions as a bearing, and rotatably supports the second human

power transmission body **612**. As shown in FIG. **4**, the one-way clutch **63** includes four balls **63D**, and the four balls **63D** are disposed at substantially regular intervals in the circumferential direction. Note that the rolling bodies are not limited to the balls **63D**, and may be members having cylindrical shapes.

Second Embodiment

[0118] Hereinafter, by reference to FIGS. **5** and **6**, a second embodiment of a drive unit in the present disclosure is described. FIG. **5** is a sectional view of a drive unit **20A** according to the second embodiment. FIG. **6** shows an enlarged view of the vicinity of a motor **40A** in FIG. **5**. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0119] As shown in FIG. **5**, the drive unit **20A** includes: the motor **40A** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31** that houses the motor **40A**; and a crankshaft **30** rotatably attached to the housing **31**. Similar to the drive unit **20** in the first embodiment, the drive unit **20A** is a so-called center unit type drive unit.

[0120] Similar to the drive unit **20** according to the first embodiment, the drive unit **20A** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40A**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50** that decelerates the rotational power of the motor **40A**, and transmits the power to the sprocket **70** via the output body **62**.

[0121] Meanwhile, as shown in FIG. **6**, the motor **40A** of the drive unit **20A** in the second embodiment is a so-called outer-rotor type motor, in which a rotor **42A** is disposed outside a stator **41A** in the radial direction. Use of the outer-rotor type facilitates outputting of higher torque than the inner-rotor type motor.

[0122] As shown in FIG. **6**, the rotor **42A**, and tooth portions **43A** of a motor shaft **43**, tooth portions **512A** of an internal gear **512**, and tooth portions **513A** of planetary gears **513** are disposed at overlapping positions in the axial direction of the crankshaft **30**. Accordingly, also in the case of using the outer-rotor type motor, the size of the drive unit **20A** can be reduced in the axial direction of the crankshaft **30**.

Third Embodiment

[0123] Next, by reference to FIGS. **7** and **8**, a third embodiment of a drive unit in the present disclosure is described. FIG. **7** is a sectional view of a drive unit **20B** according to the third embodiment. FIG. **8** shows an enlarged view of the vicinity of a motor **40** in FIG. **7**. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0124] As shown in FIG. **7**, similar to the drive unit **20** according to the first embodiment, the drive unit **20B** includes: the motor **40** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31** that houses the motor **40**; and a crankshaft **30** rotatably attached to the housing **31**.

[0125] The drive unit **20B** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50B** that decelerates the rotational power of the motor **40**, and transmits the power to the sprocket **70** via the output body **62**. The power transmission mechanism **50B** is made up of two stages of speed reduction mechanisms, and includes: a planetary gear mechanism **51** that is a first-stage speed reduction mechanism; and a second speed reduction mechanism **52** that is a second-stage speed reduction mechanism.

[0126] Meanwhile, as shown in FIG. 8, the configuration of a planetary carrier **515B** included in the planetary gear mechanism **51** is different from that of the planetary carrier **515** of the drive unit **20** in the first embodiment. Specifically, the planetary carrier **515B** includes a hollow portion **5151** centered at the rotation axis, and a protrusion member **36** and a bearing **326** are disposed in the hollow portion **5151**.

[0127] The protrusion member **36** is fitted into a concavity **3112** formed in a first divided body **311**, and is non-rotatably fixed to the first divided body **311**. No particular limitation is imposed on the material of the protrusion member **36**. However, for example, this member is made of a metal material containing iron or aluminum as a main component. By providing the protrusion member **36**, the rotation of the planetary carrier **515B** can be stabilized.

[0128] The bearing **326** rotatably supports the planetary carrier **515B** at the housing **31**. By providing the bearing **326** in the hollow portion **5151** of the planetary carrier **515B**, the size of the drive unit **20B** can be reduced.

Fourth Embodiment

[0129] Next, by reference to FIGS. 9 and 10, a fourth embodiment of a drive unit in the present disclosure is described. FIG. 9 is a sectional view of a drive unit **20C** according to the fourth embodiment. FIG. 10 shows an enlarged view of the vicinity of a motor **40** in FIG. 9. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0130] As shown in FIG. 9, similar to the drive unit **20** according to the first embodiment, the drive unit **20C** includes: the motor **40** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31** that houses the motor **40**; and a crankshaft **30** rotatably attached to the housing **31**.

[0131] The drive unit **20C** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50C** that decelerates the rotational power of the motor **40**, and transmits the power to the sprocket **70** via the output body **62**. The power transmission mechanism **50C** is made up of two stages of speed reduction mechanisms, and includes: a planetary gear mechanism **51** that is a first-stage speed reduction mechanism; and a second speed reduction mechanism **52** that is a second-stage speed reduction mechanism.

[0132] As shown in FIG. 10, the power transmission mechanism **50C** of the drive unit **20C** includes an output member **516** that meshes with a planetary carrier **515C** constituting the output shaft of the planetary gear mechanism **51**. The output member **516** is a solid cylindrical member that extends in the axial direction of the crankshaft **30**. The output member **516** is disposed such that the rotation axis of the output member **516** is positioned concentrically on the rotation axis of the planetary carrier **515C**. By providing the output member **516**, the rotation of the planetary carrier **515C** can be stabilized.

[0133] A protruding portion **5161** that protrudes outward in the radial direction from the output member **516** is formed over the entire periphery of the output member **516** at the axial-direction center portion of the output member **516**. The output member **516** and the planetary carrier **515C** mesh with each other in the left region of the protruding portion **5161**. That is, the output member **516** and the planetary carrier **515C** do not mesh with each other in the right region of the protruding portion **5161**.

[0134] The output member **516** includes a small-diameter portion **5162** formed to have a smaller diameter than the other portion, in the right region of the protruding portion **5161**. A bearing **327** (third bearing) that rotatably supports the output member **516** at the housing **31** is disposed outside of the small-diameter portion **5162** in the radial direction. Accordingly, the output member **516** can

be rotatably supported at the housing **31** without increasing the size of the drive unit **20C**. The bearing **327** (third bearing) and a first transmission gear **521** are disposed at positions overlapping in the axial direction of the crankshaft **30**. Accordingly, reduction in size of the drive unit **20C** can be facilitated.

Fifth Embodiment

[0135] Hereinafter, by reference to FIGS. **11** and **12**, a fifth embodiment of a drive unit in the present disclosure is described. FIG. **11** is a sectional view of a drive unit **20D** according to the fifth embodiment. FIG. **12** shows an enlarged view of the vicinity of a motor **40D** in FIG. **11**.

Hereinafter, components common to those in the fourth embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the fourth embodiment are mainly described.

[0136] As shown in FIG. **11**, the drive unit **20D** includes: the motor **40D** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31** that houses the motor **40A**; and a crankshaft **30** rotatably attached to the housing **31**. Similar to the drive unit **20C** in the fourth embodiment, the drive unit **20D** is a so-called center unit type drive unit.

[0137] Similar to the drive unit **20C** according to the fourth embodiment, the drive unit **20D** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40D**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50C** that decelerates the rotational power of the motor **40D**, and transmits the power to the sprocket **70** via the output body **62**.

[0138] Meanwhile, as shown in FIG. **12**, in the drive unit **20D** according to the fifth embodiment, the configuration of the motor **40D** is different from that of the drive unit **20C** according to the fourth embodiment. Specifically, the motor **40D** includes: a motor shaft **43D** that is an output shaft; a stator **41D** fixed to the housing **31**; and a rotor **42D** that is fixed to the motor shaft **43D** and includes magnets **45D**. The rotor **42D** is disposed so as to face the stator **41D** in a direction along the rotation axis of the motor shaft **43D**. That is, the motor **40D** is a so-called axial gap motor. Use of the axial gap motor can reduce the size of the motor **40D**, and facilitates reduction in the size of the drive unit **20D**, in comparison with an inner-rotor type or an outer-rotor type motor.

[0139] In the present embodiment, the rotor **42D** is disposed only on one side (left side) of the stator **41D** in the direction along the rotation axis of the motor shaft **43D**. Accordingly, the motor **40D** can be reduced in size, which facilitates reduction in the size of the drive unit **20D**. Note that the rotor **42D** may be disposed on the opposite sides of the stator **41D** in the direction along the rotation axis of the motor shaft **43D**.

[0140] The motor shaft **43D** is rotatably supported by bearings **320** and **321** at the housing **31**. The bearings **320** and **321** are disposed separately from each other in the direction along the rotation axis of the motor shaft **43D**.

[0141] As shown in FIG. **12**, the stator **41D**, and tooth portions **43A** of the motor shaft **43D**, tooth portions **512A** of an internal gear **512**, and tooth portions **513A** of planetary gears **513** are disposed at overlapping positions in the axial direction of the crankshaft **30**. Accordingly, also in the case of using the axial gap motor, the size of the drive unit **20D** can be reduced in the axial direction of the crankshaft **30**.

Sixth Embodiment

[0142] Next, by reference to FIG. **13**, a sixth embodiment of a drive unit in the present disclosure is described. FIG. **13** is a sectional view of a drive unit **20E** according to the sixth embodiment.

Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0143] As shown in FIG. **13**, the drive unit **20E** includes: a motor **40** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31E** that houses the motor **40**; and a

crankshaft **30** rotatably attached to the housing **31E**.

[0144] Similar to the drive unit **20** according to the first embodiment, the drive unit **20E** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50** that decelerates the rotational power of the motor **40**, and transmits the power to the sprocket **70** via the output body **62**.

[0145] The housing **31E** includes: a first divided body **311E** that constitutes a right outer shell of the housing **31E**; a second divided body **312E** that constitutes a left outer shell of the housing **31E**; and a third divided body **313E** that intervenes between the first divided body **311E** and the second divided body **312E**. The first divided body **311E** and the third divided body **313E** are joined with a fastener member **341**, such as a bolt, and the second divided body **312E** and the third divided body **313E** are joined with a fastener member **342**, such as a bolt. The housing **31E** is made up of these three divided bodies, which can facilitate the configuration of each divided body, and improves productivity. Note that the number of the divided bodies constituting the housing may be four or more. The space formed in the housing **31E** may be closed, or not closed.

[0146] The third divided body **313E** includes a protruding portion **3131** that protrudes to the left. An internal gear **512** included in a planetary gear mechanism **51** is fitted to the protruding portion **3131**.

Seventh Embodiment

[0147] Next, by reference to FIGS. **14** and **15**, a seventh embodiment of a drive unit in the present disclosure is described. FIG. **14** is a sectional view of a drive unit **20F** according to the seventh embodiment. FIG. **15** shows an enlarged view of the vicinity of a motor **40** in FIG. **14**. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0148] As shown in FIG. **14**, the drive unit **20F** includes: the motor **40** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31F** that houses the motor **40**; and a crankshaft **30** rotatably attached to the housing **31F**. Similar to the housing **31E** in the fifth embodiment, the housing **31F** is made up of three divided bodies: a first divided body **311F**, a second divided body **312F**, and a third divided body **313F**.

[0149] The drive unit **20F** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50F** that decelerates the rotational power of the motor **40**, and transmits the power to the sprocket **70** via the output body **62**. The power transmission mechanism **50F** is made up of two stages of speed reduction mechanisms, and includes: a planetary gear mechanism **51F** that is a first-stage speed reduction mechanism; and a second speed reduction mechanism **52** that is a second-stage speed reduction mechanism.

[0150] As shown in FIG. **15**, the planetary gear mechanism **51F** includes: a motor shaft **43** as a sun gear **511F**; an internal gear **512F** that is disposed on the concentric circle of the sun gear **511** and serves as an output shaft; planetary gears **513F** that mesh with the sun gear **511F** and the internal gear **512F**; and planetary gear shafts **514F** as fixed shafts.

[0151] The internal gear **512F** includes: an annular portion **5121** that has an annular shape, and has tooth portions that mesh with the planetary gears **513F** and are formed in the inner peripheral surface; a connection portion **5122** that is connected to the annular portion **5121** and extends inward in the radial direction; and a cylindrical body portion **5123** that is connected to the connection portion **5122** and extends along the axial direction of the crankshaft **30**. The annular portion **5121**, the connection portion **5122**, and the cylindrical body portion **5123** rotate integrally.

[0152] The rotational power of the motor **40** is transmitted to a first transmission gear **521** included in the second speed reduction mechanism **52**, via the cylindrical body portion **5123**. Note that a one-way clutch **53** is provided between the cylindrical body portion **5123** and the first transmission gear **521**.

[0153] The planetary gear shafts **514F** are respectively fitted into depressed portions **3132** formed in the third divided body **313F**. Accordingly, the planetary gear shafts **514F** are non-rotatably fixed to the housing **31F**. Note that in the example shown in FIGS. **14** and **15**, the planetary gear shafts **514F** are fixed to the third divided body **313F**. Alternatively, the planetary gear shafts **514F** may be fixed to the second divided body **312F**. The housing **31F** is made up of three divided bodies in the example shown in FIGS. **14** and **15**, but may be made up of two divided bodies, instead.

Eighth Embodiment

[0154] Next, by reference to FIGS. **16** and **17**, an eighth embodiment of a drive unit in the present disclosure is described. FIG. **16** is a sectional view of a drive unit **20G** according to the eighth embodiment. FIG. **17** shows an enlarged view of the vicinity of a motor **40** in FIG. **16**. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0155] As shown in FIG. **16**, similar to the drive unit **20** according to the first embodiment, the drive unit **20G** includes: the motor **40** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31** that houses the motor **40**; and a crankshaft **30** rotatably attached to the housing **31**.

[0156] The drive unit **20G** further includes: a human power transmission body **61** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62** that outputs the rotational power of the motor **40**; a sprocket **70** that outputs the rotational power of the output body **62**; and a power transmission mechanism **50G** that decelerates the rotational power of the motor **40**, and transmits the power to the sprocket **70** via the output body **62**.

[0157] As shown in FIG. **17**, the power transmission mechanism **50G** is a parallel shaft gear mechanism that meshes with the motor shaft **43** as the output shaft of the motor **40**, and includes a first transmission gear **541** having a rotation axis disposed in parallel with the rotation axis of the motor shaft **43**. The power transmission mechanism **50G** is a mechanism that includes the first transmission gear **541**, a gear shaft **542**, and a second transmission gear **543**, and transmits the rotational power of the motor **40** to the output body **62** via the first transmission gear **541**, the gear shaft **542**, and the second transmission gear **543**.

[0158] The first transmission gear **541** is disposed inside an opening portion of a rotor **42** in the radial direction. The first transmission gear **541** includes tooth portions **541A** that mesh with tooth portions **43A** provided on the motor shaft **43**. The rotor **42**, the tooth portions **43A** of the motor shaft **43**, and the tooth portions **541A** of the first transmission gear **541** are disposed at positions overlapping in the axial direction of the crankshaft **30**. Accordingly, the size of the drive unit **20G** can be reduced in the axial direction of the crankshaft **30**. Splines or serrations to be fitted to the gear shaft **542** are formed in the inner peripheral surface of the first transmission gear **541**. Note that the first transmission gear **541** may be subjected to insert molding on the gear shaft **542**.

[0159] The gear shaft **542** is disposed along the axial direction of the crankshaft **30**. The gear shaft **542** rotates integrally with the first transmission gear **541**. While no specific limitation is imposed on the length of the gear shaft **542**, the length ranges, for example, from 5% to 50%, inclusive, of the length of the crankshaft **30** in the axial direction. Tooth portions **542A** are formed on the right side of the gear shaft **542**.

[0160] The second transmission gear **543** meshes with the tooth portions **542A** formed on the right side of the gear shaft **542**. The rotation axis of the second transmission gear **543** is concentrically positioned on the rotation axis of the crankshaft **30**.

[0161] A one-way clutch **53G** is provided between the second transmission gear **543** and the output body **62**. That is, the rotational power of the motor **40** output from the second transmission gear **543** is transmitted to the output body **62** via the one-way clutch **53G**.

[0162] When the rotational power in the forward direction is applied to the second transmission gear **543**, the one-way clutch **53G** transmits the rotational power to the output body **62**. When the rotational power in the direction opposite to the forward direction is applied to the second transmission gear **543**, the one-way clutch **53G** does not transmit the rotational power to the output body **62**. When the rotational power in the forward direction is applied to the output body **62** via the human power transmission body **61**, the one-way clutch **53G** does not transmit the rotational power to the second transmission gear **543**.

Ninth Embodiment

[0163] Next, by reference to FIG. **18**, a ninth embodiment of a drive unit in the present disclosure is described. FIG. **18** is a sectional view of a drive unit **20H** according to the ninth embodiment. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0164] As shown in FIG. **18**, the drive unit **20H** includes: a motor **40H** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31H** that houses the motor **40H**; and a crankshaft **30** rotatably attached to the housing **31H**.

[0165] The drive unit **20H** further includes: a human power transmission body **61H** that is attached so as to rotate integrally with the crankshaft **30**, and is rotated by a human-power driving force; an output body **62H** that outputs the rotational power of the motor **40H**; a sprocket **70** that outputs the rotational power of the output body **62H**; and a power transmission mechanism **50H** that decelerates the rotational power of the motor **40H**, and transmits the power to the sprocket **70** via the output body **62H**.

[0166] The motor **40H** includes a stator **41H**, and a rotor **42H** having an opening portion at the rotation center. In the present embodiment, the motor **40H** includes no motor shaft rotating integrally with the rotor **42H**. Note that the motor **40H** may be an outer-rotor type or axial gap type motor. Depressed portions **46H** into which planetary gear shafts **504H** included in the power transmission mechanism **50H** described later are respectively fitted are formed in the motor **40H**.

[0167] Similar to the first embodiment, the housing **31H** includes: a first divided body **311H** that constitutes the right half of the housing **31H**; and a second divided body **312H** that constitutes the left half of the housing **31H**. The first divided body **311H** and the second divided body **312H** are joined together with a fastener member **34**, such as a bolt. Splines or serrations to be fitted to a first internal gear **5021H** included in the power transmission mechanism **50H** described later are formed in the inner surface of the first divided body **311H**. Note that the first divided body **311H** and the first internal gear **5021H** may be screwed together with a screw.

[0168] The power transmission mechanism **50H** is a planetary gear mechanism that includes an internal gear **502H**, planetary gears **503H**, and planetary gear shafts **504H**.

[0169] The power transmission mechanism **50H** is a mechanism that decelerates the rotational power of the motor **40H**, and transmits the power to the sprocket **70** via the output body **62H**. In the present embodiment, the planetary gear **503H** is a stepped planetary gear that includes two gears having different outer diameters.

[0170] The internal gear **502H** includes the first internal gear **5021H** and a second internal gear **5022H**. The first internal gear **5021H** meshes with a large-diameter portion of the planetary gear **503H**, and the second internal gear **5022H** meshes with a small-diameter portion of the planetary gear **503H**.

[0171] The first internal gear **5021H** is fitted to the inner surface of the first divided body **311H** constituting the housing **31H**, thus being non-rotatably fixed to the housing **31H**. That is, the first internal gear **5021H** functions as a fixed shaft of the planetary gear mechanism.

[0172] Meanwhile, the second internal gear **5022H** functions as the output shaft of the planetary gear mechanism, and the rotational power of the second internal gear **5022H** is transmitted to the output body **62H** via a one-way clutch **53H**. The second internal gear **5022H** is rotatably supported by a bearing **320H** at the housing **31H**.

[0173] As described above, the planetary gear **503H** is a stepped planetary gear that includes the two gears having different outer diameters. The planetary gear shafts **504H** penetrate through the centers of the respective planetary gears **503H**, and hold these planetary gears **503H** in a freely rotatably manner. The left end of the planetary gear shaft **504H** is fitted into the depressed portion **46H** formed in the motor **40H**.

[0174] Accordingly, the planetary gear shaft **504H** rotates integrally with the motor **40H**. Note that the planetary gear **503H** is not necessarily a stepped one, and may be made up of a single gear. In this case, the gear inner and outer diameters of the first internal gear **5021H** and the second internal gear **5022H** are substantially the same. However, by making the addendum modification coefficients in gear data different, the numbers of tooth portions of the gears can be changed. In the case of this configuration, the shape of the planetary gear **503H** is simplified, which improves the assemblability of the drive unit **20H**.

[0175] The human power transmission body **61H** is a cylindrical member that extends along the axial direction of the crankshaft **30**, and is disposed on the outer peripheral portion of the crankshaft **30** in the housing **31H**. The human power transmission body **61H** rotates integrally with the crankshaft **30**. In the present embodiment, the human power transmission body **61H** is made up of a single member. Similar to the first embodiment, it may be made up of multiple members.

[0176] The output body **62H** is a cylindrical member that extends along the axial direction of the crankshaft **30**, and is disposed on the outer peripheral portion of the crankshaft **30**. The rotation axis of the output body **62H** is concentrically positioned on the rotation axis of the crankshaft **30**. The output body **62H** is rotatably supported by bearings **321H** and **322H** at the housing **31H**. The deceleration scheme as in the drive unit **20H** in the present embodiment tends to have a low power transmission efficiency at a small number of revolutions, and generate heat. Preferably, oil lubrication for facilitating dissipation of the generated heat is adopted accordingly. The oil lubrication can reduce the friction coefficients of the tooth portions of the gears, and reduce the heat generation at the same time. Furthermore, the oil lubrication makes oil adhere to the housing **31H**, which can efficiently dissipate heat from the housing **31H**. By applying the oil on the motor **40H**, the motor **40H** can efficiently dissipate heat.

Tenth Embodiment

[0177] Next, by reference to FIG. **19**, a tenth embodiment of a drive unit in the present disclosure is described. FIG. **19** is a sectional view of a drive unit **20J** according to the tenth embodiment. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0178] As shown in FIG. **19**, the drive unit **20J** includes: a motor **40J** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **31J** that houses the motor **40J**; and a crankshaft **30** rotatably attached to the housing **31J**.

[0179] The drive unit **20J** further includes: an output body **62J** that outputs the rotational power of the motor **40J**; a sprocket **70** that outputs the rotational power of the output body **62J**; and a power transmission mechanism **50J** that decelerates the rotational power of the motor **40J**, and transmits the power to the sprocket **70** via the output body **62J**. Note that the drive unit **20J** includes no human power transmission body, and the rotational power of the crankshaft **30** is directly transmitted to the output body **62J**. Absence of the human power transmission body allows the components constituting the power transmission mechanism **50J** to be disposed closer to the crankshaft **30**, and facilitates reduction in the size of the drive unit **20J**.

[0180] The motor **40J** includes a stator **41J**, and a rotor **42J** having an opening portion at the

rotation center. In the present embodiment, the motor **40J** includes no motor shaft rotating integrally with the rotor **42J**. Note that the motor **40J** may be an outer-rotor type motor or an axial gap type motor. Depressed portions **46J** into which planetary gear shafts **504J** included in the power transmission mechanism **50J** described later are respectively fitted are formed in the motor **40J**.

[0181] Similar to the first embodiment, the housing **31J** includes: a first divided body **311J** that constitutes the right half of the housing **31J**; and a second divided body **312J** that constitutes the left half of the housing **31J**. The first divided body **311J** and the second divided body **312J** are joined together with a fastener member **34**, such as a bolt. Splines or serrations to be fitted to a first internal gear **5021J** included in the power transmission mechanism **50J** described later are formed in the inner surface of the first divided body **311J**. The first divided body **311J** includes a partition wall **3111J** that extends toward the crankshaft **30**. The partition wall **3111J** is in contact with a bearing **322J** on its outer side in the radial direction, and holds the bearing **322J**. The partition wall **3111J** may be made up of a component (e.g., an insulator) of the stator **41J**.

[0182] The inside of the housing **31J** is partitioned by the partition wall **3111J**. The motor **40J** and the power transmission mechanism **50J** are disposed in the space on the right side of the partition wall **3111J**. A control board **81J** is disposed in the space on the left side of the partition wall **3111J**. Here, the space on the right side of the partition wall **3111J** stores lubrication oil. Accordingly, the gears constituting the power transmission mechanism **50J** are lubricated, and the durability and quiet performance are improved. Note that the space in which the control board **81J** is disposed is provided with no lubrication oil.

[0183] Preferably, an elastic member, such as an O-ring, is provided between the stator **41J** and the housing **31J** (first divided body **311J**) in view of preventing leakage of the lubrication oil stored in the space on the right side of the partition wall **3111J**. Preferably, an oil seal **37J** is provided on the outer peripheral portion of the crankshaft **30**. Preferably, a bush **38J** made of metal, such as iron, is provided between the oil seal **37J** and the motor **40J**. This can prevent leakage of the lubrication oil stored in the space on the right side of the partition wall **3111J**, and reduce abrasion due to contact of a lip part of the oil seal **37J** with the motor **40J**.

[0184] The power transmission mechanism **50J** is a planetary gear mechanism that includes an internal gear **502J**, planetary gears **503J**, and planetary gear shafts **504J**. The power transmission mechanism **50J** is a mechanism that decelerates the rotational power of the motor **40J**, and transmits it to the sprocket **70** via the output body **62J**. In the present embodiment, the planetary gear **503J** is a stepped planetary gear that includes two gears having different outer diameters.

[0185] The internal gear **502J** includes a first internal gear **5021J** and a second internal gear **5022J**. The first internal gear **5021J** meshes with a large-diameter portion of the planetary gear **503J**, and the second internal gear **5022J** meshes with a small-diameter portion of the planetary gear **503J**.

[0186] The first internal gear **5021J** is fitted to the inner surface of the first divided body **311J** included in the housing **31J**, thus being non-rotatably fixed to the housing **31J**. That is, the first internal gear **5021J** functions as a fixed shaft of the planetary gear mechanism.

[0187] On the other hand, the second internal gear **5022J** functions as the output shaft of the planetary gear mechanism, and the rotational power of the second internal gear **5022J** is transmitted to the output body **62J** via a one-way clutch **53J**. The second internal gear **5022J** is rotatably supported by a bearing **320J** at the housing **31J**.

[0188] As described above, the planetary gear **503J** is a stepped planetary gear that includes the two gears having different outer diameters. The planetary gear shafts **504J** penetrate through the centers of the respective planetary gears **503J**, and hold these planetary gears **503J** in a freely rotatable manner. The left end of the planetary gear shaft **504J** is fitted into the depressed portion **46J** formed in the motor **40J**. Accordingly, the planetary gear shaft **504J** rotates integrally with the motor **40J**. The bearing provided between the planetary gear **503J** and the planetary gear shaft **504J** may be a ball bearing or a sliding bearing.

[0189] The output body **62J** is a cylindrical member that extends along the axial direction of the crankshaft **30**, and is disposed on the outer peripheral portion of the crankshaft **30**. The rotation axis of the output body **62J** is concentrically positioned on the rotation axis of the crankshaft **30**. In the present embodiment, as described above, no human power transmission body is provided. Accordingly, the rotational power of the crankshaft **30** is transmitted to the output body **62J** via the one-way clutch **63J**. The one-way clutch **63J** also functions as a bearing, and rotatably supports the output body **62J**.

[0190] The drive unit **20J** is provided with a plurality of (e.g., two to four) strain sensors **80J** as torque detecting portions. The strain sensors **80J** are disposed on the outer side of the bearing **321J** in the radial direction, and measure a load applied to the bearing **321J**. The bearing **321J** supports the crankshaft **30** on the opposite side of a site where the sprocket **70** is provided, in the axial direction of the crankshaft **30**, and is resistant to being affected by the chain tension accordingly. Thus, the strain sensors **80J** can achieve correct detection. A rotation detecting portion **82J** that detects the rotation of the crankshaft **30** is disposed on the control board **81J**. The rotation detecting portion **82J** internally includes, for example, a Hall element, and is disposed at a position overlapping a magnet **86J** provided on the outer peripheral portion of the crankshaft **30**. The Hall element detects the variation in the magnetic field of the magnet **86J** by the rotation of the crankshaft **30**, thus allowing measurement of the number of rotations of the crankshaft **30**. Note that the configuration of the rotation detecting portion **82J** is not limited to this.

Eleventh Embodiment

[0191] Next, by reference to FIGS. **20** and **21**, an eleventh embodiment of a drive unit in the present disclosure is described. FIG. **20** is a sectional view of a drive unit **120** according to the eleventh embodiment. FIG. **21** shows an enlarged view of the vicinity of a motor **140** in FIG. **20**. Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0192] As shown in FIG. **20**, the drive unit **120** includes: the motor **140** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **131** that houses the motor **140**; and a crankshaft **130** rotatably attached to the housing **131**.

[0193] The drive unit **120** includes: a human power transmission body **161** that is attached so as to rotate integrally with the crankshaft **130** and is rotated by a human-power driving force; a first output body **162** that outputs the rotational power of the motor **140**; and a second output body **163** that outputs the rotational power by the human-power driving force. The drive unit **120** includes: a first sprocket **171** that outputs the rotational power of the first output body **162**; a second sprocket **172** that outputs the rotational power of the second output body **163**; and a planetary gear mechanism **150** as a power transmission mechanism that decelerates the rotational power of the motor **140** and transmits the power to the first sprocket **171** via the first output body **162**. That is, the first sprocket **171** outputs the rotational power of the motor **140**, and the second sprocket **172** outputs the rotational power of the human-power driving force.

[0194] The motor **140** includes a stator **141**, a rotor **142**, and a motor shaft **143** that rotates integrally with the rotor **142**. In the present embodiment, the motor **140** is an inner-rotor type motor. Note that the motor **140** may be an outer-rotor type motor.

[0195] The rotor **142** has an annular shape that has a circular opening portion **144** (see FIG. **21**) at the rotation center, and is disposed inside the stator **141**. Inside the opening portion **144** in the radial direction there are a sun gear **1501**, an internal gear **1502**, planetary gears **1503**, and planetary gear shafts **1504** that are gears constituting the planetary gear mechanism **150**.

[0196] The rotor **142** and the motor shaft **143** are fixed to each other by serrations. Note that the rotor **142** and the motor shaft **143** may be fixed by press-fitting, shrink-fitting, cool-fitting, or the like. When the rotor **142** and the motor shaft **143** are fixed, an elastic member, such as any of a rubber member and urethane, may be provided between the rotor **142** and the motor shaft **143**.

[0197] The motor shaft **143** is a cylindrical shaft-shaped member extending in the left-right direction, and rotates integrally with the rotor **142**. The motor shaft **143** is disposed such that its rotation axis is along the axial direction of the crankshaft **130**. The motor shaft **143** is rotatably supported by bearings **1320**, **1321**, and **1322** provided in the housing **131**, so as to be rotatable with respect to the housing **131**.

[0198] As shown in FIG. **21**, the motor shaft **143** is disposed such that the central axis of rotation passes through the center of the internal gear **1502** included in the planetary gear mechanism **150** and penetrates through the internal gear **1502**. Tooth portions **143A** to mesh with tooth portions **1503A** of the planetary gears **1503** included in the planetary gear mechanism **150** are provided on the surface of the motor shaft **143**. The motor shaft **143** constitutes the sun gear **1501** of the planetary gear mechanism **150**.

[0199] Note that the size, shape, and the like of the motor **40** in the first embodiment may be appropriately adopted as the size, shape, and the like of the motor **140**.

[0200] The housing **131** is a member that constitutes an outer shell of the drive unit **120**, and houses the motor **140**. The housing **131** is formed mainly of metal, such as aluminum, stainless steel, or the like. Alternatively, non-metal may be used. No specific limitation is imposed on the material of the housing **131**.

[0201] The housing **131** includes: a first divided body **1311** that constitutes the right half of the housing **131**; and a second divided body **1312** that constitutes the left half of the housing **131**. The first divided body **1311** and the second divided body **1312** are joined together with a fastener member **134**, such as a bolt. By joining the first divided body **1311** and the second divided body **1312** to each other, the hollow housing **131** is formed. Note that no particular limitations are imposed on the size, shape, thickness and the like of the housing **131**. The space formed in the housing **131** may be closed, or not closed.

[0202] The first divided body **1311** includes a protruding portion **13111** that protrudes to the left. The internal gear **1502** included in the planetary gear mechanism **150** is fitted to the protruding portion **13111**.

[0203] A plurality of grooves **1312A** are formed in the wall surface of the second divided body **1312**. Thus, the area of a portion of the second divided body **1312** that is in contact with external air can be increased. As a result, the temperature of the housing **131** can be lowered, and the reliability of the drive unit **120** can be improved. Note that grooves may be formed in the wall surface of the first divided body **1311** instead of or in addition to the wall surface of the second divided body **1312**.

[0204] The inside of the housing **131** houses the human power transmission body **161**, part of the first output body **162**, part of the second output body **163**, and the planetary gear mechanism **150**, in addition to the motor **140**. The housing **131** is provided with through-holes **1314** and **1315** through which the crankshaft **130** penetrates. Furthermore, the housing **131** is provided with a through-hole **1316** through which the first output body **162** penetrates.

[0205] Note that in the present embodiment, the housing **131** is attached to the outer side of a frame **2**, but may be housed in the frame **2** instead, similar to the housing **31** in the first embodiment. In the case of housing the housing **131** in the frame **2**, a heat dissipation member that transfers heat of the housing **131** to the frame **2** may be disposed between the housing **131** and the frame **2**.

[0206] The crankshaft **130** is a cylindrical member rotationally driven by the human-power driving force. The crankshaft **130** may be made up of a hollow member, or may be made up of a solid member.

[0207] A pair of crank arms (not shown) are provided at the opposite ends of the crankshaft **130**. The opposite ends of the crankshaft **130** protrude outward from the housing **131**. The crankshaft **130** is rotatably supported by bearings **1324** and **1325** at the housing **131** (frame **2**). The bearings **1324** and **1325** are made up of, for example, ball bearings.

[0208] The planetary gear mechanism **150** is a mechanism that decelerates the rotational power of

the motor **140**, and transmits the power to the first sprocket **171** via the first output body **162**. In the present embodiment, the planetary gear mechanism **150** includes: the motor shaft **143** as the sun gear **1501**; the internal gear **1502** disposed on a concentric circle of the sun gear **1501**; the planetary gears **1503** that mesh with the sun gear **1501** and the internal gear **1502**; the planetary gear shafts **1504**; and a planetary carrier **1505** as an output shaft.

[0209] The internal gear **1512** has an annular shape. Tooth portions **1502A** are formed over the entire region of the inner peripheral surface. The internal gear **1502** is non-rotatably fixed to the housing **131** by being fitted to the protruding portion **13111** of the housing **131**.

[0210] The planetary gear **1503** is a stepped planetary gear that includes two gears having different outer diameters. The tooth portions **1503A** are formed over the entire periphery of each planetary gear **1503**. The small-diameter portion of the planetary gear **1503** meshes with the internal gear **1502**, and the large-diameter portion of the planetary gear **1503** meshes with the sun gear **1501**. Use of the stepped planetary gear **1503** facilitates an increase in the reduction ratio of the planetary gear mechanism **150**. Note that, similar to the first embodiment, the planetary gear **1503** is not necessarily the stepped planetary gear.

[0211] The planetary gear shafts **1504** penetrate through the centers of the respective planetary gears **1503**, and hold these planetary gears **1503** in a freely rotatable manner. The planetary carrier **1505** supports the planetary gears **1503** to allow them to orbit. The rotation axis of the planetary carrier **1505** is positioned concentrically on the rotation axis of the sun gear **1501**.

[0212] As shown in FIG. **21**, the rotor **142**, and the tooth portions **143A** of the motor shaft **143**, the tooth portions **1502A** of the internal gear **1502**, and the tooth portions **1503A** of the planetary gears **1503** are disposed at overlapping positions in the axial direction of the crankshaft **130**. Accordingly, in the axial direction of the crankshaft **130**, the size of the drive unit **120** can be reduced.

[0213] At least parts of the tooth portions **143A** of the motor shaft **143**, the tooth portions **1502A** of the internal gear **1502**, and the tooth portions **1503A** of the planetary gears **1503** are disposed inside the opening portion **144** of the rotor **142** in the radial direction. Accordingly, the drive unit **120** can be further reduced in size.

[0214] Note that the shapes, sizes, materials and the like of the gears of the planetary gear mechanism **51** in the first embodiment may be appropriately adopted as the shapes, sizes, materials and the like of the planetary gear mechanism **150**.

[0215] The rotational power of the motor **140** output from the planetary gear mechanism **150** is transmitted to the first output body **162** via a one-way clutch **153**. When the rotational power in the forward direction is applied to the planetary carrier **1505**, the one-way clutch **153** transmits the rotational power to the first output body **162**. When the rotational power in the direction opposite to the forward direction is applied to the planetary carrier **1505**, the one-way clutch **153** does not transmit the rotational power to the first output body **162**.

[0216] The human power transmission body **161** is a cylindrical member that extends along the axial direction of the crankshaft **130**, and is disposed on the outer peripheral portion of the crankshaft **130** in the housing **131**. The human power transmission body **161** rotates integrally with the crankshaft **130**. In the present embodiment, the human power transmission body **161** is divided into a first human power transmission body **1611** and a second human power transmission body **1612**. Note that the human power transmission body **161** may be made up of a single member.

[0217] The first human power transmission body **1611** is coupled to the crankshaft **130**. Splines or serrations to be fitted to the crankshaft **130** are formed in the inner peripheral surface of the first human power transmission body **1611**. Note that a tooth missing part may be provided at a site where the first human power transmission body **1611** and the crankshaft **130** are fitted to each other. This allows positioning in the assembling direction, and facilitates assembly. The first human power transmission body **1611** and the crankshaft **130** may be fitted together by press-fitting or with a screw.

[0218] The second human power transmission body **1612** is disposed on the right side of the first

human power transmission body **1611** in the axial direction of the crankshaft **130**. The second human power transmission body **1612** is coupled to the first human power transmission body **1611**, and transmits the rotational power to the second output body **163**.

[0219] In the present embodiment, splines or serrations to be fitted to the inner peripheral surface of the left end of the second human power transmission body **1612** are formed in the outer peripheral surface of the right end of the first human power transmission body **1611**. Accordingly, the first human power transmission body **1611** and the second human power transmission body **1612** are coupled to each other.

[0220] The first output body **162** is a shaft-shaped member which extends along the axial direction of the crankshaft **130** and to which the rotational power of the motor **140** is transmitted. The rotation axis of the first output body **162** is concentrically positioned on the rotation axis of the motor shaft **143** (sun gear **1501**). The rotational power of the motor **140** output from the planetary gear mechanism **150** is transmitted to the first output body **162** via the one-way clutch **153**.

[0221] The length of the first output body **162** in the axial direction of the crankshaft **130** is shorter than the length of the crankshaft **130**, and ranges, for example, from 10% to 50%, inclusive, of the length of the crankshaft **130** in the axial direction. The right end of the first output body **162** protrudes to the outside of the housing **131** through the through-hole **1316** provided in the housing **131**. The first output body **162** is rotatably supported by a bearing **1326** at the housing **131**.

[0222] Splines or serrations to be fitted to the first sprocket **171** are formed in a portion of the first output body **162** that protrudes to the outside of the housing **131**. Accordingly, the first sprocket **171** rotates integrally with the first output body **162**.

[0223] The second output body **163** is a cylindrical member that extends along the axial direction of the crankshaft **130**, and is disposed on the outer peripheral portion of the crankshaft **130**. The rotation axis of the second output body **163** is concentrically positioned on the rotation axis of the crankshaft **130**.

[0224] Splines or serrations to be fitted to the outer peripheral portion of the second human power transmission body **1612** are formed in the inner peripheral surface on the left end side of the second output body **163**. Accordingly, the second output body **163** rotates integrally with the second human power transmission body **1612**. That is, the rotational power by the human-power driving force output from the human power transmission body **161** is transmitted to the second output body **163**.

[0225] The length of the second output body **163** in the axial direction of the crankshaft **130** is shorter than the length of the crankshaft **130**, and ranges, for example, from 10% to 50%, inclusive, of the length of the crankshaft **130** in the axial direction. The right end of the second output body **163** protrudes to the outside of the housing **131** through the through-hole **1314** provided in the housing **131**. The second output body **163** is rotatably supported by the bearing **1324** at the housing **131**.

[0226] Splines or serrations to be fitted to the second sprocket **172** are formed in a portion of the second output body **163** that protrudes to the outside of the housing **131**. Accordingly, the second sprocket **172** rotates integrally with the second output body **163**.

[0227] The first sprocket **171** is disposed on the rear side of the second sprocket **172**. A chain (not shown) is wound around the first sprocket **171** and the second sprocket **172**. The distance between the center of the first sprocket **171** and the center of the second sprocket **172** is, for example, equal to or less than 150 mm, and may be equal to or less than 135 mm.

[0228] The second sprocket **172** has a larger outer diameter than the first sprocket **171**. The number of teeth of the second sprocket **172** is greater than the number of teeth of the first sprocket **171**.

Twelfth Embodiment

[0229] Next, by reference to FIGS. **22** and **23**, a twelfth embodiment of a drive unit in the present disclosure is described. FIG. **22** is a sectional view of a drive unit **220** according to the twelfth embodiment. FIG. **23** shows an enlarged view of the vicinity of a motor **240** in FIG. **22**.

Hereinafter, components common to those in the first embodiment are assigned the same symbols, redundant description of these components is omitted, and the differences from the first embodiment are mainly described.

[0230] As shown in FIG. 22, the drive unit **220** includes: the motor **240** for applying rotational power for assisting the pedaling force on pedals **7**; a housing **231** that constitutes part of an outer shell of the drive unit **220**; and a crankshaft **230** rotatably attached to the housing **231**. As described in detail later, the rotation axis of the motor **240** and the rotation axis of the crankshaft **230** are concentrically disposed.

[0231] The drive unit **220** further includes: a pair of crank arms **6** that have hollow structures; a human power transmission body **261** that is attached so as to rotate integrally with the crankshaft **230**, and is rotated by a human-power driving force; an output body **262** that outputs the rotational power of the motor **240**; a sprocket **270** that outputs the rotational power of the output body **262**; and a power transmission mechanism **250** that decelerates the rotational power of the motor **240**, and transmits the power to the sprocket **270** via the output body **262**. As described in detail later, the power transmission mechanism **250** is made up of two stages of speed reduction mechanisms, and includes: a first planetary gear mechanism (first power transmission mechanism) **251** that is a first-stage speed reduction mechanism; and a second planetary gear mechanism (second power transmission mechanism) **252** that is a second-stage speed reduction mechanism.

[0232] The crankshaft **230** is a cylindrical member that has a hollow structure and opens at the opposite ends. At least part of the power transmission mechanism **250** is disposed in the crankshaft **230**. Accordingly, the drive unit **220** can be further reduced in size. The crankshaft **230** is provided with a through-hole **2301** through which a planetary gear **2523**, described later, penetrates. The crankshaft **230** is rotatably supported by bearings **2320** and **2321** at the housing **231** (frame **2**). The bearings **2320** and **2321** are made up of, for example, ball bearings.

[0233] The opposite ends of the crankshaft **230** protrude outward from the housing **231**. A right crank arm **6A** is connected to the right end of the crankshaft **230**. The right crank arm **6A** includes: a base portion **6A1** that constitutes a main part of the right crank arm **6A** and has an opening portion on the left side; and a lid portion **6A2** that blocks the opening portion. The base portion **6A1** and the lid portion **6A2** are joined to each other with a fastener member **6A3**, such as a bolt. By joining the base portion **6A1** and the lid portion **6A2** together, the hollow right crank arm **6A** is formed.

[0234] The internal space of the crankshaft **230** and the internal space of the right crank arm **6A** communicate with each other. The left end side of the crankshaft **230** penetrates through the left crank arm **6B**, and a power storage device **10** is attached to the left end of the crankshaft **230**. Although described in detail later, the power storage device **10** has a configuration such that it is attachable to and detachable from the left end of the crankshaft **230**.

[0235] Splines or serrations to be fitted to an internal gear **2512** included in the first planetary gear mechanism **251**, described later, are formed in the inner peripheral surface of the crankshaft **230** on its right end side.

[0236] The housing **231** is a cylindrical member that extends along the axial direction of the crankshaft **230** and covers the center portion of the crankshaft **230**. The housing **231** is fixed in, for example, a bottom bracket (not shown). Parts of the human power transmission body **261** and the output body **262** are housed in the housing.

[0237] Splines or serrations to be fitted to an internal gear **2522** included in the second planetary gear mechanism **252**, described later, are formed in the inner peripheral surface of the housing **231** on its right end side.

[0238] As shown in FIG. 23, the motor **240** includes a stator **241**, a rotor **242**, and a motor shaft **243** that rotates integrally with the rotor **242**. In the present embodiment, the motor **240** is an inner-rotor type motor. Note that the motor **240** may be an outer-rotor type or axial gap type motor.

[0239] The motor **240** is housed in the crank arm **6** (right crank arm **6A**) and in the crankshaft **230**.

Specifically, the stator **241** and the rotor **242** are housed in the right crank arm **6A**, and the motor shaft **243** is housed in the crankshaft **230**. By housing the motor **240** in the crank arms **6** and in the crankshaft **230**, the size of the drive unit **220** can be reduced.

[0240] The rotor **242** has an annular shape that has a circular opening portion **244** at the rotation center, and is disposed inside the stator **241**. Inside the opening portion **244** in the radial direction there are parts of a sun gear **2511**, the internal gear **2512**, planetary gears **2513**, and a planetary gear shaft **2514** that are gears constituting the first planetary gear mechanism **251**.

[0241] The rotor **242** and the motor shaft **243** are fixed to each other by serrations. Note that the rotor **242** and the motor shaft **243** may be fixed by press-fitting, shrink-fitting, cool-fitting, or the like. When the rotor **242** and the motor shaft **243** are fixed, an elastic member, such as any of a rubber member and urethane, may be provided between the rotor **242** and the motor shaft **243**.

[0242] The motor shaft **243** is a cylindrical shaft-shaped member extending in the left-right direction, and rotates integrally with the rotor **242**. The motor shaft **243** is disposed such that its rotation axis is along the axial direction of the crankshaft **230**. The motor shaft **243** is rotatably supported by a bearing **2322** provided in the right crank arm **6A** with respect to the crankshaft **230** and the right crank arm **6A**.

[0243] The motor shaft **243** is disposed such that the central axis of rotation passes through the center of the internal gear **2512** included in the planetary gear mechanism **251**, and penetrates through the internal gear **2512**. Tooth portions **243A** to mesh with tooth portions **2513A** of the planetary gears **2513** included in the first planetary gear mechanism **251** are provided on the surface of the motor shaft **243**. That is, the motor shaft **243** functions as the sun gear **2511** of the first planetary gear mechanism **251**, and constitutes a gear of the first planetary gear mechanism **251**.

[0244] Note that the size, shape, and the like of the motor **40** in the first embodiment may be appropriately adopted as the size, shape, and the like of the motor **240**.

[0245] The power transmission mechanism **250** is a mechanism that decelerates the rotational power of the motor **240**, and transmits the power to the sprocket **270** via the output body **262**. In the present embodiment, the power transmission mechanism **250** is made up of two stages of speed reduction mechanisms, and includes: the first planetary gear mechanism **251** that is a first-stage speed reduction mechanism; and the second planetary gear mechanism **252** that is a second-stage speed reduction mechanism. Note that the power transmission mechanism **250** may be made up of a single-stage speed reduction mechanism.

[0246] As shown in FIG. **23**, the first planetary gear mechanism **251** includes: the motor shaft **243** as the sun gear **2511**; the internal gear **2512** disposed on the concentric circle of the sun gear **2511**; the planetary gears **2513** that mesh with the sun gear **2511** and the internal gear **2512**; the planetary gear shafts **2514**; and a planetary carrier **2515** as an output shaft.

[0247] The internal gear **2512** has an annular shape. Tooth portions **2512A** are formed over the entire region of the inner peripheral surface. The internal gear **2512** is non-rotatably fixed to the crankshaft **230** by being fitted to the inner peripheral surface of the crankshaft **230**.

[0248] The tooth portions **2513A** are formed over the entire periphery of each planetary gear **2513**. The planetary gear shafts **2514** penetrate through the centers of the respective planetary gears **2513**, and hold these planetary gears **2513** in a freely rotatably manner. The planetary carrier **2515** supports the planetary gears **2513** to allow them to orbit. The rotation axis of the planetary carrier **2515** is positioned concentrically on the rotation axis of the sun gear **2511**.

[0249] As shown in FIG. **23**, the rotor **242**, and the tooth portions **243A** of the motor shaft **243**, the tooth portions **2512A** of the internal gear **2512**, and the tooth portions **2513A** of the planetary gears **2513** are disposed at overlapping positions in the axial direction of the crankshaft **230**. Accordingly, in the axial direction of the crankshaft **230**, the size of the drive unit **220** can be reduced.

[0250] At least parts of the tooth portions **243A** of the motor shaft **243**, the tooth portions **2512A** of the internal gear **2512**, and the tooth portions **2513A** of the planetary gears **2513** are disposed inside

the opening portion **244** of the rotor **242** in the radial direction. Accordingly, the drive unit **220** can be further reduced in size.

[0251] The rotational power of the motor **240** output from the first planetary gear mechanism **251** is transmitted to the second planetary gear mechanism **252** via a one-way clutch **253**. When the rotational power in the forward direction is applied to the planetary carrier **2515**, the one-way clutch **253** transmits the rotational power to the sun gear **2521** as an input shaft of the second planetary gear mechanism **252**. When the rotational power in the direction opposite to the forward direction is applied to the planetary carrier **2515**, the one-way clutch **253** does not transmit the rotational power to the sun gear **2521**.

[0252] As shown in FIG. **23**, the second planetary gear mechanism **252** includes: the sun gear **2521** as an input shaft; the internal gear **2522** disposed on the concentric circle of the sun gear **2521**; the planetary gears **2523** that mesh with the sun gear **2521** and the internal gear **2522**; and planetary gear shafts **2524**.

[0253] As for the sun gear **2521**, the rotation axis of the sun gear **2521** is concentrically positioned on the rotation axis of the crankshaft **230**. As described above, the rotational power of the motor **240** output from the first planetary gear mechanism **251** is transmitted to the sun gear **2521** via the one-way clutch **253**. The sun gear **2521** is rotatably supported by the bearing **2323** provided on the left end side with respect to the crankshaft **230**.

[0254] The internal gear **2522** has an annular shape. Tooth portions **2522A** are formed over the entire region of the inner peripheral surface. The internal gear **2522** is non-rotatably fixed to the housing **231** by being fitted to the inner peripheral surface of the housing **231**.

[0255] The planetary gear **2523** is a stepped planetary gear that includes two gears having different outer diameters. The tooth portions **2523A** are formed over the entire periphery of each planetary gear **2523**. The small-diameter portion of the planetary gear **2523** meshes with the internal gear **2522**, and the large-diameter portion of the planetary gear **2523** meshes with the sun gear **2521**. Use of the stepped planetary gear **2523** facilitates an increase in the reduction ratio of the second planetary gear mechanism **252**. Note that similar to the first embodiment, the planetary gear **2523** is not necessarily the stepped planetary gear.

[0256] The planetary gear shafts **2524** penetrate through the centers of the respective planetary gears **2523**, and hold these planetary gears **2523** in a freely rotatable manner. The planetary gear shafts **2524** are inserted into through-holes formed in the output body **262**, and are fitted thereinto. Accordingly, the rotational power of the motor **240** output from the second planetary gear mechanism **252** is transmitted to the output body **262**.

[0257] Note that the shapes, sizes, materials, and the like of the gears of the planetary gear mechanism **51** in the first embodiment may be appropriately adopted as the shapes, sizes, materials, and the like of the gears of the first planetary gear mechanism **251** and the second planetary gear mechanism **252**.

[0258] As shown in FIG. **22**, the human power transmission body **261** is a cylindrical member that extends along the axial direction of the crankshaft **230**, and is disposed on the outer peripheral portion of the crankshaft **230** in the housing **231**. The human power transmission body **261** rotates integrally with the crankshaft **230**. In the present embodiment, the human power transmission body **261** is made up of a single member. Similar to the first embodiment, it may be made up of multiple members.

[0259] The human power transmission body **261** is coupled to the crankshaft **230**. Splines or serrations to be fitted to the crankshaft **230** are formed in the inner peripheral surface of the human power transmission body **261** on its left end side. The human power transmission body **261** and the crankshaft **230** may be fitted together by press-fitting or with a screw. The length of the human power transmission body **261** in the axial direction of the crankshaft **230** is shorter than the length of the crankshaft **230**, and ranges, for example, from 10% to 50%, inclusive, of the length of the crankshaft **230** in the axial direction. Note that the drive unit **220** does not necessarily include the

human power transmission body **261**. In this case, the output body **262** and the crankshaft **230** are engaged together with splines or the like, and the rotational power of the crankshaft **230** is transmitted to the output body **262**. In the case without the human power transmission body **261**, the strain of the crankshaft **230** may be detected by a strain sensor pasted to the outer peripheral surface of the crankshaft **230**, and the input human power torque may thus be detected. The input human power torque may be detected by detecting the load on the outer wheel side of a bearing **2321** through the strain sensor. This negates the need for the human power transmission body **261**, and further facilitates reduction in the size and weight of the drive unit **220**.

[0260] Tooth portions to mesh with tooth portions provided on the inner peripheral surface of the output body **262** are provided on the outer peripheral portion on the right end side of the human power transmission body **261**. Accordingly, the rotational power by the human-power driving force is transmitted to the output body **262** via the human power transmission body **261**. Note that a one-way clutch may be provided between the human power transmission body **261** and the output body **262**.

[0261] The output body **262** is a cylindrical member that extends along the axial direction of the crankshaft **230**, and is disposed on the outer peripheral portion of the crankshaft **230**. The rotation axis of the output body **262** is concentrically positioned on the rotation axis of the crankshaft **230**.

[0262] The length of the output body **262** in the axial direction of the crankshaft **230** is shorter than the length of the crankshaft **230**, and ranges, for example, from 10% to 50%, inclusive, of the length of the crankshaft **230** in the axial direction. The output body **262** is rotatably supported by the bearing **2320** at the housing **231**.

[0263] The right end of the output body **262** protrudes to the outside of the housing **231** through a through-hole **2311** provided in the housing **231**. Splines or serrations to be fitted to the sprocket **270** are formed in a portion of the output body **262** that protrudes to the outside of the housing **231**. Accordingly, the sprocket **270** rotates integrally with the output body **262**.

[0264] Note that the human power transmission body **261** and the output body **262** are separately provided in the present embodiment, but may be integrally configured. For example, the human power transmission body **261** is not necessarily provided, and it may be the case that only the output body **262** is provided. In this case, by transmitting the rotational power of the crankshaft **230** by the human-power driving force to the output body **262**, the power-assisted bicycle **1** is allowed to travel forward.

[0265] A one-way clutch may be disposed between the output body **262** and the sprocket **270**. When the rotational power in the forward direction is applied to the output body **262**, the one-way clutch transmits the rotational power to the sprocket **270**. When the rotational power in the direction opposite to the forward direction is applied to the output body **262**, the one-way clutch does not transmit the rotational power to the sprocket **270**.

[0266] As shown in FIG. **22**, the drive unit **220** includes torque detecting portions **280** that detect the torque of the human-power driving force; and a control board **281** on which a control device (not shown) for controlling the output of the motor **240** in response to the detected torque and achieving an appropriate power assisting function is disposed. The drive unit **220** further includes a rotation detecting portion **282** for detecting the rotation of the crankshaft **230**.

[0267] The torque detecting portions **280** are provided on the outer peripheral portion of the human power transmission body **261**, and detect the strain applied to the human power transmission body **261**. For example, a strain sensor that detects a physical strain may be used as each torque detecting portion **280**. For example, a scheme may be used according to which a paint that emits light in response to a mechanical stimulus is applied to the outer peripheral portion of the human power transmission body **261** and the light is detected by an optical sensor.

[0268] In the case of detecting the torque using the strain sensor, the drive unit **220** includes a total of two torque detecting portions **280**, one at each of the opposite positions in the circumferential direction on the outer peripheral portion of the human power transmission body **261**. Note that the

number of torque detecting portions **280** provided on the outer peripheral portion of the human power transmission body **261** may be only one, or three or more. The installation positions of the torque detecting portions **280** are not limited to the outer peripheral portion of the human power transmission body **261**, and may be, for example, on the outer peripheral portion of the crankshaft **230**.

[0269] The control device is disposed on the control board **281**, and controls the rotation of the motor **240** based on the torque detected by each torque detecting portion **280**. The control device controls the rotation of the motor **240** based on detection information on the rotation detecting portion **282**. A conventionally publicly known configuration may be adopted as the configuration of the control device. The torque detecting portions **280** may communicate with the control device using signal lines or wireless signals. Note that the control board **281** on which the control device is disposed may be disposed adjacent to the motor **240** as shown in FIG. **22**. For example, a rotor rotation detecting portion for detecting the rotation of the rotor **242** may be provided on the control board **281** disposed adjacent to the motor **240**.

[0270] In the present embodiment, the control board **281** is housed in the right crank arm **6A**. That is, the control board **281** rotates integrally with the crankshaft **230**. The control board **281** is housed in the right crank arm **6A**, which can facilitate reduction in size of the drive unit **220**.

[0271] The rotation detecting portion **282** detects the number of rotations of the crankshaft **230**. The rotation detecting portion **282** is, for example, an inertial sensor, such as an acceleration sensor or a gyroscope sensor. Note that the configuration of the rotation detecting portion **282** is not limited to this.

[0272] In the present embodiment, at least part of the rotation detecting portion **282** is disposed on the control board **281**. That is, at least part of the rotation detecting portion **282** rotates integrally with the crankshaft **230**. Accordingly, reduction in size of the drive unit **220** can be facilitated. Note that at least part of the rotation detecting portion **282** may be disposed on the crankshaft **230** or the human power transmission body **261**.

[0273] The rotation detecting portion **282** may communicate with the control device using signal lines and preferably using wireless signals. This can omit the space that houses a signal line connecting the rotation detecting portion **282** and the control device to each other, and facilitates reduction in the size of the drive unit **220**.

[0274] As shown in FIG. **22**, the power storage device **10** is attached to the left end of the crankshaft **230**. That is, the power storage device **10** rotates integrally with the crankshaft **230**.

[0275] The power storage device **10** includes: a housing body **10A** that houses a storage battery **10C** and is disposed outside the crankshaft **230**; and a connector **10B** connected to the housing body **10A**. The housing body **10A** and the connector **10B** are joined to each other with a fastener member **10D**, such as a bolt. The housing body **10A** includes a depressed portion **10E** into which the crankshaft **230** is fitted. A control board **10H** that controls the power storage device **10** is provided in the housing body **10A**.

[0276] The power storage device **10** has a configuration such that it is attachable to and detachable from the crankshaft **230**. Specifically, a lock mechanism **10F** is provided on the outside of the housing body **10A** in the radial direction of the rotation axis of the crankshaft **230**. When the lock mechanism **10F** is unlocked, a protruding portion **10G** is retracted from an engagement site, and the fastening at the depressed portion **10E** is loosened, and the power storage device **10** can be removed from the crankshaft **230**. By locking the lock mechanism **10F** in a state in which the crankshaft **230** is fitted into the depressed portion **10E**, the protruding portion **10G** is fitted to the engagement site, the crankshaft **230** is fastened by the depressed portion **10E**, and the power storage device **10** is fixed to the crankshaft **230**. Note that no specific limitation is imposed on the configuration of the lock mechanism **10F** so long as the configuration allows the power storage device **10** to be attached to and detached from the crankshaft **230**. For example, the lock mechanism **10F** may be disposed in the axial direction of the crankshaft **230**.

[0277] The connector **10B** is disposed in the crankshaft **230**, and is configured to be insertable into the left end of an internal connector **284** in the crankshaft **230**. It is configured such that the outer diameter of the connector **10B** is smaller than the inner diameter of the crankshaft **230**. A cable **283** that supplies electric power to the components constituting the drive unit **220**, such as the motor **240**, is connected to the right end of the internal connector **284**. In the example in FIG. 22, the cable **283** passes through the crankshaft **230**, and is connected to the control board **281** disposed in the right crank arm **6A**.

[0278] Note that the method of supplying electric power to the components constituting the drive unit **220** is not limited to this. For example, the power storage device **10** may be attached to the frame **2**, and the electric power may be supplied to the drive unit **220** using a slip ring. For example, the drive unit **220** includes: an electric power receiving portion that rotates integrally with the crankshaft **230**; and an electric power supplying portion that is fixed to the frame **2** and includes a brush for supplying electric power to the electric power receiving portion. The electric power receiving portion is provided with the slip ring. The distal end portion of the brush is in contact and slid by rotation of the slip ring, and electric power is supplied from the brush to the slip ring. Note that the method of supplying electric power using the slip ring is not limited to the form described above.

[0279] Note that the design of each of the embodiments described above may be appropriately changed in a range without changing the advantage of the present disclosure. For example, in each of the embodiments, the power-assisted bicycle **1** provided with the crank arms **6** at the opposite ends of the crankshaft **30**, **130**, **230** is described. Alternatively, this may apply to an electric vehicle without the crank arms **6** at the opposite ends of the crankshaft **30**, **130**, **230**. That is, the vehicle may be an electric vehicle that does not travel forward by a human-power driving force, but travels forward by the rotational power of the motor.

REFERENCE SIGNS LIST

[0280] **1** Power-assisted bicycle; **2** Frame; **2A** Head tube; **2B** Front forks; **2C** Top tube; **2D** Bottom tube; **2E** Seat tube; **2F** Seat stays; **2G** Chain stays; **3A** Front wheel; **3B** Rear wheel; **4** Handlebars; **5** Saddle; **6** Crank arm; **6A** Right crank arm; **6A1** Base portion; **6A2** Lid portion; **6A3** Fastener member; **6B** Left crank arm; **7** Pedals; **8** Chain; **10** Power storage device; **10A** Housing body; **10B** Connector; **10C** Storage battery; **10D** Fastener member; **10E** Depressed portion; **10F** Lock mechanism; **10G** Protruding portion; **10H** Control board; **11** Rear wheel sprocket; **20**, **20A**, **20B**, **20C**, **20D**, **20E**, **20F**, **20G**, **120**, **220** Drive unit; **30**, **130**, **230** Crankshaft; **31**, **31E**, **31F**, **131**, **231** Housing; **33** Heat dissipation member; **34**, **134**, **341**, **342** Fastener member; **35** Heat dissipation member; **36** Protrusion member; **40**, **40A**, **40D**, **140**, **240** Motor; **41**, **41A**, **41D**, **141**, **241** Stator; **42**, **42A**, **42D**, **142**, **242** Rotor; **43**, **43D**, **143**, **243** Motor shaft; **43A**, **143A**, **243A**, **512A**, **513A**, **541A**, **542A**, **1502A**, **1503A**, **2512A**, **2513A**, **2522A**, **2523A** Tooth portions; **44**, **144**, **244** Opening portion; **50**, **50B**, **50C**, **50F**, **50G**, **250** Power transmission mechanism; **51**, **51B**, **51F**, **150**, **251** Planetary gear mechanism; **52** Second speed reduction mechanism; **53**, **53G**, **63**, **153**, **253** One-way clutch; **61**, **161**, **162** Human power transmission body; **62**, **262** Output body; **70**, **270** Sprocket; **80**, **280** Torque detecting portion; **81**, **281** Control board; **82**, **282** Rotation detecting portion; **83**, **283** Cable; **84** Connector; **85** Rotor rotation detecting portion; **86**, **86J** Magnet; **162** First output body; **163** Second output body; **171** First sprocket; **172** Second sprocket; **251** First planetary gear mechanism; **252** Second planetary gear mechanism; **284** Internal connector; **311**, **311E**, **311F**, **1311** First divided body; **312**, **312E**, **312F**, **1312** Second divided body; **313E**, **313F** Third divided body; **314**, **315**, **1314**, **1315**, **1316**, **2301**, **2311** Through-hole; **320**, **321**, **322**, **323**, **324**, **325**, **326**, **327**, **1329**, **1321**, **1322**, **1324**, **1325**, **1326**, **2320**, **2321**, **2322**, **2323** Bearing; **511**, **511F**, **1501**, **2511**, **2521** Sun gear; **512**, **512F**, **1502**, **2512**, **2522** Internal gear; **513**, **513F**, **1503**, **2513**, **2523** Planetary gear; **514**, **514F**, **1504**, **2514**, **2524** Planetary gear shaft; **515**, **515B**, **515C**, **515F**, **1505**, **2515** Planetary carrier; **516** Output member; **521**, **541** First transmission gear; **522**, **543** Second transmission gear; **542** Gear shaft; **611**, **1611** First human power transmission body; **612**, **1612** Second human power

transmission body; **841** First portion; **842** Second portion; **1312A** Grooves; **3111**, **3131**, **13111** Protruding portion; **3112**, **3132** Depressed portion; **5121** Annular portion; **5122** Connection portion; **5123** Cylindrical body portion; **5151** Hollow portion; **5161** Protruding portion; **5162** Small-diameter portion

Claims

1. A drive unit for use in an electric vehicle, comprising: a motor that includes a rotor having an opening portion at a rotation center; a housing that houses the motor; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit the power to the sprocket, wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and at least part of the gear is disposed inside the opening portion.
2. A drive unit for use in an electric vehicle, comprising: a motor that includes a rotor having an opening portion at a rotation center; a housing that houses the motor; a crankshaft that is rotatably attached to the housing, and is rotatable by a human-power driving force; a first sprocket that outputs a rotational power of the motor; a second sprocket that outputs a rotational power of the human-power driving force; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit the power to the first sprocket, wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and at least part of the gear is disposed inside the opening portion.
3. The drive unit according to claim 1, wherein the crankshaft is rotatable by a human-power driving force, and the drive unit further comprises: a torque detecting portion that detects the human-power driving force applied to the crankshaft; and a control device that controls rotation of the motor, based on detection information on the torque detecting portion.
4. The drive unit according to claim 1, wherein the power transmission mechanism includes a planetary gear mechanism that includes planetary gears, planetary gear shafts, and a planetary carrier.
5. The drive unit according to claim 4, wherein the gears of the planetary gear mechanism include a helical gear.
6. The drive unit according to claim 4, wherein the planetary gear mechanism further includes an internal gear, and the internal gear is non-rotatably fixed to the housing.
7. The drive unit according to claim 4, wherein the planetary gear mechanism further includes a sun gear, and the sun gear is non-rotatably fixed to the housing.
8. The drive unit according to claim 4, wherein the planetary carrier is non-rotatably fixed to the housing.
9. The drive unit according to claim 1, wherein the power transmission mechanism is a mechanism that decelerates or accelerates the rotational power of the motor in multiple stages, and includes: a first power transmission mechanism; and a second power transmission mechanism which is on a subsequent-stage side of the first power transmission mechanism on a transmission path and to which a rotational power of the first power transmission mechanism is transmitted, the power transmission mechanism further includes: a first bearing that rotatably supports gears included in the second power transmission mechanism, at the housing; and a one-way clutch provided between the first power transmission mechanism and the second power transmission mechanism, and the first bearing and the one-way clutch are disposed at positions overlapping in the axial direction of the crankshaft.
10. The drive unit according to claim 1, wherein the power transmission mechanism decelerates or

accelerates the rotational power of the motor in multiple stages, the power transmission mechanism includes: a first power transmission mechanism that includes a sun gear, planetary gears, a planetary carrier, and an internal gear; and a second power transmission mechanism which is on a subsequent-stage side of the first power transmission mechanism on a transmission path and to which a rotational power of the first power transmission mechanism is transmitted, a second bearing that rotatably supports the sun gear at the housing is provided at one end of the sun gear in the housing, the second bearing and the second power transmission mechanism are disposed at positions overlapping in the axial direction of the crankshaft, and a rotation axis of the rotor and a rotation axis of the crankshaft are respectively disposed on different axes.

11. The drive unit according to claim 5, wherein the power transmission mechanism includes a cylindrical output member that meshes with an output shaft of the planetary gear mechanism, the output member includes a small-diameter portion formed to have a smaller diameter than other portions, and a third bearing that rotatably supports the output member at the housing is disposed on an outer side of the small-diameter portion in a radial direction.

12. The drive unit according to claim 1, wherein the power transmission mechanism includes a parallel shaft gear mechanism having a gear that meshes with a motor shaft as an output shaft of the motor and has a rotation axis disposed in parallel with a rotation axis of the motor shaft.

13. The drive unit according to claim 1, further comprising a control board on which a control device that controls rotation of the motor is disposed, wherein the control board is joined to the motor.

14. The drive unit according to claim 1, wherein the motor includes a stator, and the stator has an outer diameter equal to or smaller than 100 mm.

15. The drive unit according to claim 1, wherein the motor includes a stator, and the stator has an outer diameter equal to or smaller than 90 mm.

16. The drive unit according to claim 1, wherein the rotor has an inner diameter equal to or greater than 30 mm.

17. The drive unit according to claim 1, wherein the rotor has an inner diameter equal to or greater than 40 mm.

18. The drive unit according to claim 3, further comprising a rotation detecting portion that detects rotation of the crankshaft, wherein at least part of the rotation detecting portion is disposed on a board on which the control device is disposed.

19. The drive unit according to claim 18, wherein the torque detecting portion is disposed on the board.

20. The drive unit according to claim 3, wherein a cable extending from an outside of the drive unit is connected via a connector to the board on which the control device is disposed, and the connector includes a first portion that extends in a substantially normal direction of a surface of the board, and a second portion that extends in a direction inclined from the substantially normal direction of the surface of the board.

21. The drive unit according to claim 3, wherein the torque detecting portion rotates integrally with the crankshaft, and the torque detecting portion and the control device communicate with each other by a wireless signal.

22. The drive unit according to claim 1, wherein the housing is fixed to a frame of the electric vehicle, and the housing includes a heat dissipating portion that transfers heat to the frame.

23. The drive unit according to claim 1, wherein an interior of the housing is partitioned into at least two spaces, and a space which is one of the spaces and in which the power transmission mechanism is disposed houses lubrication oil.

24. A drive unit for use in an electric vehicle, comprising: a motor that includes a rotor having an opening portion at a rotation center; a housing; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit the power to the sprocket,

- wherein the power transmission mechanism includes at least one gear provided with tooth portions on a surface, the rotor and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft, and a rotation axis of the rotor and a rotation axis of the crankshaft are concentrically disposed.
- 25.** The drive unit according to claim 24, wherein the crankshaft has a hollow structure, and at least part of the power transmission mechanism is housed in the crankshaft.
- 26.** The drive unit according to claim 24, further comprising a power storage device that supplies electric power to the motor, wherein the power storage device rotates integrally with the crankshaft.
- 27.** The drive unit according to claim 24, further comprising a one-way clutch disposed between the crankshaft and the power transmission mechanism, wherein the one-way clutch receives a radial load.
- 28.** The drive unit according to claim 27, wherein the one-way clutch is disposed closer to the sprocket in the axial direction of the crankshaft than an axial-direction center portion of the crankshaft.
- 29.** A drive unit for use in an electric vehicle, comprising: a motor; a housing that houses the motor; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit the power to the sprocket; and a rotation detecting portion that detects rotation of the crankshaft or of a member rotating integrally with the crankshaft, and wherein the rotation detecting portion rotates integrally with the crankshaft.
- 30.** The drive unit according to claim 29, further comprising a control device that controls a rotation of the motor, based on detection information on the rotation detecting portion, wherein the rotation detecting portion and the control device communicate with each other by a wireless signal.
- 31.** The drive unit according to claim 30, wherein the rotation detecting portion and the control device are disposed on the same board.
- 32.** A drive unit for use in an electric vehicle, comprising: a motor; a housing that houses the motor; a crankshaft rotatably attached to the housing; a sprocket that outputs a rotational power of the motor; and a power transmission mechanism that decelerates or accelerates the rotational power of the motor to transmit the power to the sprocket, wherein the motor includes: a motor shaft that is an output shaft; a stator fixed to the housing; and a rotor that is fixed to the motor shaft and includes a magnet, the rotor is disposed so as to face the stator in a direction along a rotation axis of the motor shaft, the power transmission mechanism includes at least one gear provided with tooth portions on a surface, and the stator and the tooth portions are disposed at positions overlapping in an axial direction of the crankshaft.
- 33.** The drive unit according to claim 32, wherein the rotor is disposed only on one side of the stator in the direction along the rotation axis of the motor shaft.
- 34.** The drive unit according to claim 32, further comprising a plurality of bearings that rotatably support the motor shaft at the housing, wherein the plurality of bearings are separately disposed in the direction along the rotation axis of the motor shaft.
- 35.** The drive unit according to claim 1, further comprising a bearing that rotatably supports the crankshaft, and a torque detecting portion detects a load applied to the bearing.
- 36.** An electric vehicle, comprising the drive unit according to claim 1.
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